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DOCKING SLEEVE WITH ELECTRICAL ADAPTER

Abstract

A protective arrangement for an electronic device includes a flexible cover having a panel and a skirt that form an interior cavity to receive an electronic device; and an adapter fixedly positioned in the flexible cover and having a male plug with connectors extending into the interior cavity of the flexible cover for mating with a female socket of the device and a contactor with contacts adjacent outwardly from the flexible cover and electrically coupled to one or more of the connectors of the plug. A docking cradle or external adapter can receive the electronic device and cover.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application is a continuation of U.S. patent application Ser. No. 18/924,756, filed Oct. 23, 2024, which is a continuation U.S. patent application Ser. No. 17/689,728, filed Mar. 8, 2022, which issued as U.S. Pat. No. 12,132,511, which is a continuation of U.S. patent application Ser. No. 16/831,101, filed Mar. 26, 2020, which issued as U.S. Pat. No. 11,476,884, which is a continuation of U.S. patent application Ser. No. 16/233,662, filed Dec. 27, 2018, which issued as U.S. Pat. No. 10,630,334, which is a continuation of U.S. patent application Ser. No. 15/960,274, filed Apr. 23, 2018, which issued as U.S. Pat. No. 10,389,399, which is a continuation of U.S. patent application Ser. No. 15/195,176, filed Jun. 28, 2016, which issued as U.S. Pat. No. 10,050,658, which is a continuation-in-part of U.S. patent application Ser. No. 14/936,517 filed Nov. 9, 2015, which issued as U.S. Pat. No. 9,706,026, which is a continuation-in-part of U.S. patent application Ser. No. 14/829,378 filed Aug. 18, 2015, which issued as U.S. Pat. No. 9,602,639, which is a continuation-in-part of U.S. patent application Ser. No. 14/754,492, filed Jun. 29, 2015, which issued as U.S. Pat. No. 9,529,387, which is a continuation-in-part of PCT Patent Application No. PCT/US2015/017131, filed Feb. 23, 2015, which claims the benefit of U.S. Provisional Patent Application No. 62/040,037, filed Aug. 21, 2014. U.S. patent application Ser. No. 15/960,274, filed Apr. 23, 2018, which issued as U.S. Pat. No. 10,389,399, is also a continuation-in-part of U.S. patent application Ser. No. 14/222,320, filed Mar. 21, 2014, which issued as U.S. Pat. No. 9,331,444, which claims the benefit of U.S. Provisional Patent Application Ser. No. 61/943,986, filed Feb. 24, 2014, all of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to a cover for protecting a portable electronic device, and in particular to a flexible cover having an electrical adapter for coupling the device to a docking station.

BACKGROUND OF THE INVENTION

[0003] Protective covers, or 'skins', are generally well-known for protecting a portable electronic device, such as a smartphone, or tablet or another portable electronic device. Such skins are typically somewhat flexible which allows them to be wrapped around to partially envelop a device.

[0004] However, known protective covers, or 'skins', are limited in their ability to provide efficient and reliable usage of such portable electronic devices.

SUMMARY OF THE INVENTION

[0005] The present invention is a cover for protecting a portable electronic device that overcomes limitations of the prior art for efficient and reliable usage of such portable electronic devices.

[0006] One aspect of the invention is a protective arrangement for an electronic device that includes a flexible cover having a panel and a skirt surrounding the panel, where the panel and skirt form an interior cavity therebetween, and the skirt forming a mouth opening that communicates with the interior cavity that is configured and arranged to receive an electronic device; an adapter fixedly positioned in the flexible cover and having a male plug with connectors extending into the interior cavity of the flexible cover in an arrangement for mating with a female socket of the device and a contactor with contacts adjacent outwardly from the flexible cover and electrically coupled to one or more of the connectors of the plug; and a hard shell configured and arranged to be disposed around at least a portion of the flexible protective cover and having an opening through which the contactor of the adapter is extendable. The hard shell may also include corner openings and the flexible cover may include corner portions configured and arranged to protrude from the corner openings of the hard shell. The hard shell may include at least one button opening and the flexible cover may include at least one soft button configured and arranged to protrude from the at least one button opening of the hard shell.

[0007] Another aspect of the invention is a protective arrangement for an electronic device that includes a hard shell configured and arranged to be disposed around at least a portion of the electronic device and having an opening to allow access to an input/output socket of the electronic device; a flexible cover having a panel and a skirt surrounding the panel, where the panel and skirt form an interior cavity therebetween, and the skirt forming a mouth opening that communicates with the interior cavity that is configured and arranged to receive the electronic device disposed in the hard shell; and an adapter fixedly positioned in the flexible cover and including a male plug having connectors extending into the interior cavity of the flexible cover in an arrangement for mating with the input/output socket of the device through the opening in the hard shell and a contactor having contacts adjacent outwardly from the flexible cover and electrically coupled to one or more of the connectors of the plug.

[0008] Yet another aspect of the invention is a docking system that includes one of the protective arrangement described above and a docking cradle having a tray configured to receive the protective arrangement and a docking connector having contacts positioned to connect with one or more of the contacts of the contactor.

[0009] Other aspects of the invention are detailed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The foregoing aspects and many of the attendant advantages of this invention will become more readily appreciated as the same becomes better understood by reference to the following

detailed description, when taken in conjunction with the accompanying drawings, wherein:

[0011] FIG. 1 and FIG. 2 each illustrate the same invention embodied by example and without limitation as a completely integral one-piece elastomeric protective cover, or skin, in situ as partially enveloping a target portable electronic device, such as a smartphone or tablet or another portable electronic device of the prior art having a port for a docking connector;

[0012] FIG. 3, FIG. 4 and FIG. 5 describe by example and without limitation a docking cradle of a type useful with the protective cover of the invention;

[0013] FIG. 6 and FIG. 7 are different views that each illustrate by example and without limitation the protective cover of FIG. 1 with the portable device removed for clarity;

[0014] FIG. 8 is a partial cross-section view of the protective cover of FIG. 1 taken through an electrical adapter of the invention;

[0015] FIG. 9 illustrates the elastomeric protective cover of FIG. 1 in situ as partially enveloping the target portable electronic device;

[0016] FIG. 10 illustrates a plurality of electrical connectors of a male plug of the electrical adapter being mechanically and electrically mated with a female input/output socket of the target electronic device when the target device is received within an interior cavity of the flexible protective shell of the invention, as illustrated in FIG. 9;

[0017] FIG. 11 illustrates the target portable electronic device being received within the flexible protective shell of the protective cover of FIG. 1, and the protected target portable electronic device being subsequently inserted into the docking cradle;

[0018] FIG. 12 illustrates the flexible protective shell of the protective cover of FIG. 1, wherein at least a flexible center panel of the protective shell is an integral window panel formed of a substantially optically transparent flexible membrane for use with a portable electronic device having a pressure-sensitive touch screen;

[0019] FIG. 13 illustrates the portable electronic device being received within the flexible protective shell of the protective cover of FIG. 1, wherein the touch-sensitive screen of the target electronic device is positioned in contact with the integral transparent window panel of the center panel as disclosed in FIG. 12, and the protected portable electronic device being subsequently inserted into the docking cradle with the touch-sensitive screen facing outwardly away therefrom;

[0020] FIG. 14 is a close-up section view of the protective cover of FIG. 1 as illustrated in FIG. 12, wherein the protective shell is formed with the interior transparent window panel for use with the portable electronic device having a touch-sensitive screen, which illustrates the male plug of the electrical adapter being interfaced with electrical contacts of the female I/O socket of the portable electronic device;

[0021] FIG. 15 illustrates the protective cover of FIG. 1 being formed with the interior transparent window panel for use with the portable electronic device having a touch-sensitive screen as illustrated in FIG. 12, wherein the interior window panel is positioned for being juxtaposed with the touch-sensitive screen of the portable electronic device, and wherein the mouth opening through the side skirt of the protective shell is positioned adjacent to one side thereof between the transparent window panel and an opposing panel of the protective shell;

[0022] FIG. 16 is a block diagram that illustrates the electrical adapter of the protective cover of FIG. 1 being configured as a pass-through electrical adapter having electrical couplings between sequentially ordered pairs of the plurality of electrical connectors of its male plug and corresponding electrical contacts of its contactor;

[0023] FIG. 17 is a block diagram that illustrates an alternative embodiment wherein the electrical adapter of the protective cover of FIG. 1 is configured as a converting electrical adapter having electrical couplings between non-sequentially ordered pairs of the plurality of electrical connectors of its male plug and corresponding electrical contacts of its contactor;

[0024] FIG. 18 illustrates the electrical adapter of the protective cover of FIG. 1 having the converting adapter configured as its male plug and contactor separated into two individual adapter

plug and contactor bodies which are located in different areas of the protective shell;

[0025] FIG. **19** and FIG. **20** are cross-section views of the protective cover of FIG. **1**, wherein each of FIG. **19** and FIG. **20** illustrate the individual plug and contactor bodies of the electrical adapter are interconnected in the protective shell of the protective cover, for example by elongated electrical cables, wires or other electrical conductors;

[0026] FIG. **21** illustrates the converting adapter configured as being separated into a plurality of male plugs (two shown) each located in the side skirt of the protective cover of FIG. **1** at different areas of elastomer shell;

[0027] FIG. **22** illustrates the protective cover of FIG. **1** having the male plug of the converting adapter separate and spaced apart from the contactor of the adapter, as more clearly illustrated in subsequent figures;

[0028] FIG. **23** illustrates the protective cover of FIG. **1** having the contactor of the converting adapter provided on external surface of the flexible center panel, while the docking cradle is alternatively configured to position the docking connector thereof in a position for coupling with the contactor thus positioned;

[0029] FIG. **24** illustrates one embodiment of the docking cradle configured with a female nest for receiving the protective cover of FIG. **1** having the contactor of the converting adapter provided on external surface of the flexible center panel, as illustrated in FIG. **25**, and wherein the docking cradle includes one part (magnets shown) of a magnetic attraction couple operable between the protective cover and the docking cradle for securing the protective cover in the docking cradle;

[0030] FIG. **25** illustrates the unitary protective shell of the protective cover of FIG. **1** formed with the locator dam configured as a complementary male nesting appendage configured to seat in female nest of the docking cradle embodiment illustrated in FIG. **24**, and wherein the protective cover includes one part (magnetically permeable ferromagnetic member shown) of the magnetic attraction couple operable between the protective cover and the docking cradle for securing the protective cover in the docking cradle;

[0031] FIG. **26** is a cross-section view that illustrates the elastomeric shell of the protective cover of FIG. **1** as illustrated in FIG. **25** being seated in the tray of the docking cradle embodiment illustrated in FIG. **24**, wherein the complementary male nesting appendage of the protective cover is nested with the female nest of the docking cradle, wherein the portion of the magnetic attraction couple resident in each of the protective cover and the docking cradle is operable for securing the shell of the protective cover in the docking cradle;

[0032] FIG. **27**, FIG. **28** and FIG. **29** together illustrate an alternative embodiment of the protective cover of FIG. **1** as illustrated in FIG. **25**, wherein the components of the magnetic attraction couple are inverted between the protective cover and the docking cradle, i.e., wherein the magnetically permeable members are substituted for the magnets in the docking cradle, while the magnets are substituted for the magnetically permeable members in the protective cover, wherein FIG. **27** illustrates the docking cradle configured with the magnetically permeable members, FIG. **28** illustrates the protective cover configured with the magnets, and FIG. **29** is a cross-section view that illustrates the protective cover of FIG. **1** as illustrated in FIG. **28** being seated in the tray of the docking cradle embodiment illustrated in FIG. **27**;

[0033] FIG. **30**, FIG. **31** and FIG. **32** together illustrate an alternative embodiment of cooperating the protective cover of FIG. **1** as illustrated in FIG. **25** and the docking cradle of FIG. **24**, wherein the positioning interface features the protective cover of FIG. **1** as illustrated in FIG. **28** formed by the male nesting appendage and mating female nest is inverted, wherein FIG. **30** illustrates the protective cover of FIG. **25** being configured with the female nest positioning interface feature, FIG. **31** illustrates the docking cradle of FIG. **24** being configured with the complementary male nesting positioning interface feature, and FIG. **32** is a cross-section view that illustrates the protective cover of FIG. **1** as illustrated in FIG. **30** being mated with the tray of the docking cradle embodiment illustrated in FIG. **31**;

[0034] FIG. 33 illustrates the protective cover of FIG. 1 as illustrated in FIG. 18 through FIG. 20, wherein the complementary male nesting appendage is configured as the positioning interface dam in one side skirt of the protective shell, as similarly shown in FIG. 6, for being received into the female nest of the base receiver of the docking cradle, as similarly shown in FIG. 11, and wherein in contrast, the positioning interface dam on the exterior of the protective shell is spaced away from the male plug of the converting adapter, which plug is extended into the protective shell at a position in a different side skirt of the shell away from the contactor and contacts of the converting adapter, as disclosed herein, and wherein the complementary male nesting appendage is further configured with one or more magnets of the magnetic attraction couple;

[0035] FIG. 34 illustrates the protective cover of FIG. 1 as illustrated in FIG. 33 having the electronic device installed therein, wherein the magnetically permeable members are substituted for the magnets of the magnetic attraction couple;

[0036] FIG. 35 is a cross-section taken through the protective cover of FIG. 1 as illustrated in FIG. 33, wherein the male plug and contactor of the electrical adapter are spaced apart in different areas of the side skirt of the protective shell, and wherein the complementary male nesting appendage is further configured with one or more magnets of the magnetic attraction couple;

[0037] FIG. 36 is a cross-section taken through the protective cover of FIG. 1 as illustrated in FIG. 34, wherein the male plug and contactor of the electrical adapter are spaced apart in different areas of the side skirt of the protective shell, and wherein the magnetically permeable members are substituted for the magnets of the magnetic attraction couple;

[0038] FIG. 37 illustrates the female nest feature of the base receiver in the docking cradle, which includes one or more ferromagnetic members, for example, exposed in flanking positions on either side of the biasing electrical contacts of the docking connector;

[0039] FIG. 38 is a cross-section view taken through FIG. 37, which illustrates the female nest features of the alternative base receiver, wherein one or more magnets are substituted for the ferromagnetic members in flanking positions on either side of the biasing electrical contacts of the docking connector, as shown in FIG. 37;

[0040] FIG. 39 is a side view illustrating one embodiment of a contact arrangement of the adapter of a cover;

[0041] FIG. 40 is a perspective view of one embodiment of an adapter for use in a cover;

[0042] FIG. 41 is a side view illustrating another embodiment of a contact arrangement of the adapter of a cover;

[0043] FIG. 42 is a perspective view of a cover that includes a peripheral cavity;

[0044] FIG. 43 is a perspective view of the cover of FIG. 42 from a different angle;

[0045] FIG. 44 is a perspective view of the cover of FIG. 42 with a peripheral device disposed in the peripheral cavity;

[0046] FIG. 45 is a perspective view of the cover of FIG. 42 with a portable electronic device disposed in the interior cavity;

[0047] FIG. 46 is a perspective view showing the back side of the cover of FIG. 42 with a peripheral device in the peripheral cavity and a portable electronic device in the interior cavity of the cover;

[0048] FIG. 47 is a perspective view of another docking cradle;

[0049] FIG. 48 is a perspective view of a stand-alone docking cradle with a portable electronic device, such as a mobile phone, in a cover and disposed on the cradle;

[0050] FIG. 49 is a perspective view of a wall mounted docking cradle with a portable electronic device, such as a tablet, in a cover and disposed on the cradle; and

[0051] FIG. 50A is a perspective front view of a protective arrangement with a flexible cover and a hard shell;

[0052] FIG. 50B is a perspective back view of the protective arrangement of FIG. 50A;

[0053] FIG. 51A is a perspective front view of one embodiment of a portable electronic device, a

cover, and an external adapter;

[0054] FIG. 51B is a perspective view of the portable electronic device, cover, and external adapter with the external adapter uncoupled from the cover;

[0055] FIG. 51C is a close-up perspective view of a portion of the electronic device, cover, and external adapter of FIGS. 51A and 51B;

[0056] FIG. 51D is a close-up perspective view of a portion of the electronic device, cover, and external adapter of FIGS. 51A and 51B with a portion of the external adapter cut away;

[0057] FIG. 52A is a perspective front view of one embodiment of a docking cradle with a movable arm in an extended position;

[0058] FIG. 52B is a cut-away side view of the docking cradle of FIG. 52A;

[0059] FIG. 52C is a perspective front view of the docking cradle of FIG. 52A with the moveable arm in the close position;

[0060] FIG. 52D is a cut-away side view of the docking cradle of FIG. 52C;

[0061] FIG. 52E is a perspective front view of the docking cradle of FIG. 52A with a portable electronic device and a cover received by the docking cradle;

[0062] FIG. 52F is a cut-away side view of the docking cradle of FIG. 52E;

[0063] FIG. 53 is a perspective view of yet another docking cradle;

[0064] FIG. 54A is a perspective view of a docking cradle that can accommodate multiple electronic devices and covers;

[0065] FIG. 54B is a perspective view of the docking cradle of FIG. 54A with multiple electronic devices and covers docked;

[0066] FIG. 55A is a perspective view of another embodiment of an adapter for use in a cover;

[0067] FIG. 55B is a perspective view of one embodiment of a cover with the adapter of FIG. 55A;

[0068] FIG. 56A is a perspective view of one embodiment of a cover with a peripheral cavity;

[0069] FIG. 56B is a perspective view of the cover of 56A with a portion of the cover cut-away;

[0070] FIG. 57A is a perspective view of an adapter for use in a cover, such as the cover of FIG. 56A, with a peripheral cavity; and

[0071] FIG. 57B is a perspective view of the adapter of FIG. 57A from a different angle.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENT

[0072] As required, a detailed illustrative embodiment of the present protective enclosure is disclosed herein. However, techniques, systems and operating structures in accordance with the present protective enclosure may be embodied in a wide variety of forms and modes, some of which may be quite different from those in the disclosed embodiment. Consequently, the specific structural and functional details disclosed herein are merely representative, yet in that regard, they are deemed to afford the best embodiment for purposes of disclosure and to provide a basis for the claims herein which define the scope of the present protective enclosure. The following presents a detailed description of an illustrative embodiment (as well as some alternative embodiments) of the present protective enclosure.

[0073] In the Figures, like numerals indicate like elements.

[0074] FIG. 1 and FIG. 2 illustrate a completely integral one-piece elastomeric protective cover, or skin, **100** of the present invention in situ as partially enveloping the portable electronic device **1** such as a smartphone or tablet or another portable electronic device of the prior art having a port for a docking connector **3**. Cover **100** of the present invention is a sheath molded of a suitable elastic or flexibly resilient elastomer, such as but not limited to vinyl, in a size and shape to fit over and closely conform to the smartphone, tablet or other portable electronic device **1** so that the cover **100** fits the device **1** like a surgical glove.

[0075] A smartphone, or smart phone, is a mobile phone with more advanced computing capability and connectivity than basic feature phones. Early smartphones typically combined the features of a mobile phone with those of another popular consumer device, such as a personal digital assistant (PDA), a media player, a digital camera, and/or a GPS navigation unit. Modern smartphones

include all of those features plus usually include the additional features of a touch-screen computer, including web browsing, Wi-Fi, and 3rd-party apps and accessories. The most popular smartphones today are powered by Google's Android and Apple's iOS mobile operating systems. See, e.g., Wikipedia, The Free Encyclopedia.

[0076] A tablet computer, or simply tablet, is a mobile computer with display, circuitry and battery in a single unit. Tablets are equipped with sensors, including cameras, microphone, accelerometer and touch-screen, with finger or stylus gestures replacing computer mouse and keyboard. Tablets may include physical buttons, e.g., to control basic features such as speaker volume and power and ports for network communications and to charge the battery. An on-screen, pop-up virtual keyboard is usually used for typing. Tablets are typically larger than smart phones or personal digital assistants at 7 inches (18 cm) or larger, measured diagonally. One example of a tablet is the iPad tablet computer from Apple. See, e.g., Wikipedia, The Free Encyclopedia.

[0077] Hybrid tablets having detachable keyboards have been sold since the mid-1990s.

Convertible touch-screen notebook computers have an integrated keyboard that can be hidden by a swivel or slide joint. Booklet tablets have dual-touch-screens and can be used as a notebook by displaying a virtual keyboard on one of the displays. See, e.g., Wikipedia, The Free Encyclopedia.

[0078] Other portable electronic devices can include, for example, testing equipment (for example, ohmmeters, multimeters, alignment tools, battery testers, calibrators, distance meters, light meters, pressure meters, radiation detectors, thermometers, or the like), cameras, media players, or the like. In at least some embodiments, the portable electronic device includes a sensor for detecting, for example, temperature, pressure, leveling, angles, altitude, light, density, x-rays, magnetic fields, acceleration, speed, velocity, distance, energy, current or energy flow, force, durometer, mass/weight, torque, resistance, capacitance, voltage, reactance, or the like or any combination thereof.

[0079] As disclosed herein, an electrical adapter is part of protective cover **100**, the adapter includes a male plug having a plurality of electrical connectors extending into a cavity formed by protective cover **100** in an arrangement for mating with a female input/output socket of the smartphone or tablet or other portable electronic device **1**, and a contactor having a plurality of electrical contacts that are positioned adjacent to an exterior of protective cover **100** and are electrically coupled to one or more of the connectors of the plug.

[0080] Accordingly, when the electronic device **1** is received within the cavity of flexible protective cover **100**, the plurality of connectors of the male plug are mated with the female socket of the device **1**. Thereafter, the female input/output socket is electrically accessed through the plurality of contacts of the contactor that are exterior of protective cover **100**.

[0081] Protective cover **100**, with the electronic device **1** installed therein, is then inserted into a docking cradle **5** which includes docking connector **3** having its own plurality of contacts. The docking connector is joined to the contactor of the electrical adapter with one or more of the plurality of contacts of the docking connector being electrically coupled with contacts of the contactor. Accordingly, the input/output socket of the electronic device **1** is electrically accessed through the docking connector of the docking cradle **5** via the electrical adapter of protective cover **100**.

[0082] FIG. **3**, FIG. **4** and FIG. **5** describe by example and without limitation a docking cradle **5** of a type useful with protective cover **100**. For example, docking cradle **5** has a tray **7** with a base receiver **9** sized to receive a specific smartphone, tablet or other portable electronic device **1** already protected within the cover **100**. The tray **7** includes a back support surface **101** to support a back of the cover **100** and a docking support surface **103** extending away from the back support surface. The back support surface **101** and docking support surface **103** may form an angle in a range of 90 to 130 degrees or more.

[0083] Docking connector **3** is provided in the docking cradle **5**, often in the middle of base receiver **9** of tray **7**. The docking connector **3** may be specific to a particular portable electronic

device **1**, or may be generic to a type of device **1**. Docking connector **3** includes a plurality of biasing electrical contacts **11**, typically biasing pogo pins or biasing leaf spring contacts. Electrical contacts **11** are coupled to leads **13** in an electrical wire or cable **15** coupled to docking cradle **5**. Docking connector **3** may be a female connector as illustrated in FIGS. **3-5** and may define a rim **105**. The rim **105** may be recessed with respect to the docking support surface **103**.

[0084] Docking connector **3** may also include optional guide pins **17** adapted for mating with mating guide pin receivers in the contactor of the electrical adapter, as disclosed herein.

[0085] Docking cradle **5** optionally includes a clamp **19** adapted for capturing the specific portable electronic device **1** within base receiver **9** of tray **7**. Clamp **19** also has a compression component, i.e. arm **21** that operates to compress the electrical contacts of the contactor on the exterior of protective cover **100** into contact with electrical contacts **11** of docking connector **3**.

[0086] FIG. **6** and FIG. **7** illustrate the protective cover **100** of the invention with the portable electronic device **1** removed for clarity. Protective cover **100** is complimentary in shape to portable electronic device **1**. Protective cover **100** is shown being formed of a flexible unitary protective shell **102** having a flexible center panel **104** surrounded by an integral flexible side skirt **106**. Panel **104** and skirt **106** combine to form an interior cavity **108** therebetween that is sized to at least partially cover the side faces and one of the front and back faces of portable electronic device **1**, and to further extend around a peripheral edge of an opposite one of the front and back faces of portable electronic device **1**. A continuous integral lip **110** extends inward of side skirt **106** and forms a mouth opening **112** that communicates with cavity **108**. Mouth opening **112** is sized to receive electronic device **1** therethrough into cavity **108**.

[0087] Protective cover **100** includes an adapter **114** that is fixedly positioned in side skirt **106**. For example, adapter **114** is either over molded, bonded or fastened through side skirt **106**. Adapter **114** is formed of a male plug **116** having a plurality of electrical connectors **118** that extend into cavity **108** of shell **102** in an arrangement for mating with a female input/output socket of the portable electronic device **1**. Adapter **114** also includes a contactor **120** having a plurality of electrical contacts **122** that are spaced away from male plug **116** at an opposite end of adapter **114**, as illustrated in FIGS. **8** and **9**. Adapter **114** is extended through protective shell **102**, for example, through either center panel **104** or side skirt **106** (shown). Thus, while male plug **116** is extended into cavity **108** of shell **102**, contactor **120** and its contacts **122** are positioned adjacent to an exterior **124** of shell **102**, whereby plurality of electrical contacts are exposed for connection to electrical contacts **11** of docking connector **3**.

[0088] FIG. **8** is a partial cross-section view of protective cover **100** taken through electrical adapter **114**. As illustrated here, electrical contacts **122** of contactor **120** are electrically coupled to one or more of electrical connectors **118** of plug **116** via electrical cables, wires, traces or other electrical conductors **126** (one shown for clarity) embedded in a body **128** of adapter **114**.

Contactor **120** of adapter **114** is optionally formed with one or more guide pin receivers **130** sized and positioned to mate with optional guide pins **17**, if present, of docking connector **3**.

[0089] Optionally, contactor **120** of adapter **114** is further recessed in outer surface **124** of unitary protective shell **102**. By example and without limitation, an optional dam **132** is formed in outer surface **124** of unitary protective shell **102**, and adapter contactor **120** is further recessed within dam **132**. Electrical contacts **122** of contactor **120** are protected from damage by being recessed within dam **132** or otherwise within outer surface **124** of unitary protective shell **102**.

[0090] FIG. **9** illustrates the elastomeric protective cover **100** of the present invention in situ as partially enveloping the portable electronic device **1**. Here, portable electronic device **1** is received within cavity **108** of flexible protective shell **102** through mouth opening **112** that communicates with cavity **108**. Electronic device **1** is inserted into flexible protective shell **102** through mouth opening **112** in continuous integral lip **110** through side skirt **106** that communicates with cavity **108**. One of either a front face **23** (shown) or a back face of electronic device **1** is exposed in mouth opening **112**. Here, a touch-sensitive screen or other display **25** of portable electronic device **1** is

exposed in mouth opening **112** for easy operation by user.

[0091] FIG. **10** illustrates plurality of electrical connectors **118** of male plug **116** are mechanically and electrically mated with the female input/output socket **27** of device **1** when portable electronic device **1** is received within cavity **108** of flexible protective shell **102**, as illustrated in FIG. **9**. Accordingly, mating of male plug **116** with female input/output socket **27** creates access to power, sensor and/or other functions of portable electronic device **1** from outside surface **124** of unitary protective shell **102**.

[0092] FIG. **11** illustrates portable electronic device **1** being received within flexible protective shell **102** of protective cover **100**, and protective cover **100** with portable electronic device **1** therein being subsequently inserted into docking cradle **5**. Docking connector **3** of docking cradle **5** is joined to contactor **120** of electrical adapter **114** with one or more of plurality of biasing electrical contacts **11** of docking connector **3** being electrically coupled with electrical contacts **122** of adapter contactor **120**. Accordingly, portable electronic device **1** is accessed by docking cradle **5** without being first removed from protective cover or 'skin' **100** as required with prior art protective covers. Compression of clamp **19**, when present, aids in maintaining electrical coupling between electrical contacts **122** of adapter contactor **120** and biasing electrical contacts **11** of docking connector **3**.

[0093] FIG. **12** illustrates protective cover **100** reversed on electronic device **1** with center panel **104** adjacent to the face of device **1** having display **25**. Protective shell **102** is formed with mouth opening **112** that communicates with cavity **108** positioned substantially opposite from center panel **104**. Continuous integral lip **110** extends inward of side skirt **106** and forms mouth opening **112**.

[0094] Here, an interior window opening **134** is formed in flexible center panel **104** of flexible protective shell **102** opposite from mouth opening **112** and in a position over touch-sensitive screen display **25** of portable electronic device **1** for operation of electronic device **1**.

[0095] Center panel **104** of protective shell **102** is formed with interior window opening **134** that communicates with cavity **108**. Interior window opening **134** is positioned and sized to reveal therethrough pressure-sensitive touch screen **25** of a portable electronic device **1** residing in cavity **108**. A continuous integral lip **136** extends around periphery of center panel **104** inward of side skirt **106** and forms interior window opening **134**.

[0096] An integral interior window panel **138** is positioned in flexible center panel **104** within window opening **134**, and is retained in protective shell **102** by continuous integral lip **136** surrounding opening **134**. For example, as more clearly illustrated in FIG. **14**, a peripheral edge **140** of interior window panel **138** is welded, adhered, bonded or otherwise joined to integral lip **136** of window opening **134**. Optionally, a substantially watertight joint **142** is formed between interior window panel **138** and integral lip **136** of window opening **134**. According to one embodiment, interior window panel **138** is overmolded into integral lip **136** of window opening **134** for joining to integral lip **136** and forming watertight joint **142** therebetween.

[0097] Integral interior window panel **138** is formed of a substantially optically transparent membrane for use with portable electronic device **1** having a pressure-sensitive touch screen **25**, whereby operation of electronic device **1** via touch-sensitive screen **25** is accomplished through intervening window panel **138**.

[0098] Optionally, interior window panel **138** is formed of a hard, inelastic and noncompressible material. For example, the plastic sheet component forming interior window panel **138** is optionally a noncompressible sheet of relatively hard and substantially visually transparent material, including but not limited to polyethylene terephthalate (PET) or another suitable substantially visually transparent material. According to one embodiment, the plastic sheet component forming interior window panel **138** is optionally a noncompressible sheet of relatively hard and substantially visually transparent polyvinyl chloride (PVC) material. As disclosed herein, interior window panel **138** formed of PET or PVC or other hard, noncompressible and inelastic plastic sheet material similarly permits tactile stimulation of the touch-sensitive screen display **25** of portable electronic

device **1**.

[0099] Alternatively, optionally either the entirety of protective shell **102**, or only interior window panel **138**, is formed of an optically transparent flexible elastomer membrane which is either clear, or may be tinted with color, such as pink or blue as may be aesthetically pleasing. In other embodiments, the protective shell **102** is opaque or translucent and may be any color or pattern. In such embodiments, the portable electronic device **1** is disposed within the protective cover **100** with its display visible through the mouth opening **112**.

[0100] Optionally, the optically transparent elastomer of protective shell **102** or only interior window panel **138** is a dielectric material having static dissipative properties, such as an effective dielectric constant on the order of about 4.5. For example, the elastomer includes a quantity of powdered metal, mixed-metal oxides, polymers, or is coated with a static dissipative coating. Optionally, the optically transparent elastomer of protective shell **102** or only interior window panel **138** is polarized to reduce glare. Accordingly, protective cover **100** is optionally utilized with portable electronic device **1** having a pressure-sensitive touch screen that permits inputting data of and operating the device **1** by application of light pressure by the user, as is generally well-known.

[0101] Portable electronic device **1** is illustrated as being inserted through second mouth opening **112** into cavity **108** of flexible protective shell **102** with touch-sensitive screen **25** being positioned in contact with interior window panel **138** of opposing center panel **104**, whereby the flexibility of the elastomeric material permits application of pressure to touch-sensitive screen **25** for inputting data and operating the electronic device **1** in general. As disclosed herein, interior window panel **138** of center panel **104** is structured as a thin flexible membrane in a position that corresponds to touch-sensitive screen **25** of device **1**. By example and without limitation, window area of flexible center panel **104** is expected to be about 0.05 to 0.06 inch thick, or another thickness suitable for permitting touch-sensitive screen **25** to respond to pressure exerted by the user for inputting data and operating the electronic device **1** in general. Integral interior window panel **138** of center panel **104** covers pressure-sensitive touch screen **25** of electronic device **1** to protect it from scratching, denting, and other external abrasions. Side skirt **106** and portions of protective shell **102** surrounding interior window panel **138** of center panel **104** may be thicker for providing protection from bumps and scrapes.

[0102] Male plug **116** of adapter **114** is inserted into female input/output socket **27** of electronic device **1**.

[0103] FIG. **13** illustrates portable electronic device **1** being received within flexible protective shell **102** of protective cover **100** with touch-sensitive screen **25** being positioned in contact with interior window panel **138**. Protective cover **100** with portable electronic device **1** therein is subsequently inserted into docking cradle **5** with touch-sensitive screen **25** facing outwardly away therefrom. Docking connector **3** of docking cradle **5** is joined to contactor **120** of electrical adapter **114** with one or more of plurality of biasing electrical contacts **11** of docking connector **3** being electrically coupled with electrical contacts **122** of adapter contactor **120**. Accordingly, portable electronic device **1** is accessed by docking cradle **5** without being first removed from protective cover or 'skin' **100** as required with prior art protective covers. When present, compression of clamp **19** aids in maintaining electrical coupling between electrical contacts **122** of adapter contactor **120** and biasing electrical contacts **11** of docking connector **3**.

[0104] FIG. **14** is a close-up section view of protective cover **100** as illustrated in FIG. **12**, wherein protective shell **102** is formed with interior window panel **138**. FIG. **14** illustrates male plug **116** of electrical adapter **114** being interfaced with plurality of electrical contacts **29** of female input/output socket **27** of the portable electronic device **1**.

[0105] FIG. **15** illustrates protective cover **100** being formed with interior transparent window panel **138** in a position for being juxtaposed with touch-sensitive screen **25** of portable electronic device **1**. However, mouth opening **112** through side skirt **106** is positioned adjacent to one side **144** of unitary protective shell **102** between interior window panel **138** in center panel **104** and an

opposing panel **146** of protective shell **102**. Here, mouth opening **112** is formed by continuous integral lip **110** that extends inward of side skirt **106** at side **144** of protective shell **102**.

[0106] When mouth opening **112** is formed through side skirt **106** adjacent to one side **144** of protective shell **102**, electrical adapter **114** is positioned in side skirt **106** at a second side **148** opposite from side **144**. With mouth opening **112** and electrical adapter **114** thus located at opposite sides **144** and **148**, respectively, of protective shell **102**, male plug **116** of electrical adapter **114** is extended into cavity **108** of shell **102** in a position to be received into and mechanically and electrically mated with the female input/output socket **27** of device **1** when portable electronic device **1** is received within cavity **108** through side mouth opening **112**. Accordingly, in practice portable electronic device **1** is inserted (arrow **150**) into protective shell **102** through side mouth opening **112** and into interior cavity **108** until the female input/output socket **27** is mechanically mated with male plug **116** of electrical adapter **114**, whereupon female input/output socket **27** is simultaneously electrically mated with electrical adapter **114**. Portable electronic device **1** is removed from protective shell **102** in reverse order along the insertion direction (arrow **150**).

[0107] FIG. **16** is a block diagram that illustrates adapter **114** being further configured as a pass-through adapter having electrical couplings between sequentially ordered pairs of the plurality of electrical connectors **118** of male plug **116** and corresponding electrical contacts **122** of contactor **120**. Here, wires or other electrical conductors **126** embedded in body **128** of adapter **114** communicate in direct sequential relationship between electrical connectors **118** of male plug **116** and corresponding electrical contacts **122** of contactor **120**. When adapter **114** is configured as a pass-through adapter, and portable electronic device **1** (shown in phantom) is received into protective cover **100** and the two received into docking cradle **5** (shown in phantom), as shown, input/output signals on a plurality of electrical contacts **29** of female socket **27** of the portable electronic device **1** are presented to electrical contacts **11** of docking connector **3** in the same order as presented on electrical contacts **29** of female input/output socket **27** of portable electronic device **1**. Accordingly, electrical contacts **11** of docking connector **3** must be ordered correspondingly to electrical contacts **29** of input/output socket **27** of electronic device **1**. Any desired reordering of input/output signals on electrical contacts **29** of female socket **27** of the electronic device **1** must occur in docking connector **3** of docking cradle **5**, or elsewhere downstream of docking connector **3**.

[0108] FIG. **17** is a block diagram that illustrates an alternative embodiment wherein adapter **114** is instead further configured as a converting adapter having electrical couplings between non-sequentially ordered pairs of the plurality of electrical connectors **118** of male plug **116** and corresponding electrical contacts **122** of contactor **120**. Here, wires, traces or other electrical conductors **126** embedded in body **128** of adapter **114** communicate in non-sequential relationship between electrical connectors **118** of male plug **116** and electrical contacts **122** of contactor **120**. When adapter **114** is configured as a converting adapter, and portable electronic device **1** (shown in phantom) is received into protective cover **100** and the two received into docking cradle **5** (shown in phantom), as shown, input/output signals on the plurality of electrical contacts **29** of female socket **27** of the portable electronic device **1** are presented to electrical contacts **11** of docking connector **3** in any desired order, in contrast to only the order presented on electrical contacts **29** of female input/output socket **27** of portable electronic device **1** as in the pass-through configuration of adapter **114** shown in FIG. **14**. Electrical contacts **11** of docking connector **3** are thus ordered correspondingly to a predetermined input/output sequence on leads **13** in electrical cable **15** of docking cradle **5**. Accordingly, input/output signals on electrical contacts **29** of female socket **27** of the electronic device **1** are reordered in any desired sequence to match docking connector **3**, whereby downstream reordering of input/output signals on electrical contacts **29** of female socket **27** of the electronic device **1** in docking connector **3** is not required to match the specific electronic device **1** received in docking cradle **5**. Instead, docking connector **3** operates as a universal docking connector for cooperating with any device **1** inserted into docking cradle **5** when converting adapter

114 is matched to the portable electronic device **1**. Accordingly, docking cradle **5** can be provided with a single generic or universal docking connector **3** and still be utilized with a variety of different ones of the portable electronic device **1** by simply coordinating the order of coupling relationships between electrical connectors **118** of male plug **116** and electrical contacts **122** of contactor **120** within electrical adapter **114** by routing of wires, electrical traces or other electrical conductors **126** embedded in a body **128** of adapter **114**.

[0109] FIG. **18** illustrates converting adapter **114** configured as having male plug **116** and contactor **120** separated into two individual adapter plug body **128a** and contactor body **128b** which are located in different areas of protective shell **102**. Individual adapter plug body **128a** of male plug **116** and contactor body **128b** of contactor **120** are fixedly positioned in side skirt **106** of protective shell **102**. Here for example, individual adapter plug and contactor bodies **128a**, **128b** are optionally embedded and overmolded in protective shell **102**. By example and without limitation, male plug **116** is optionally embedded in skirt **106** adjacent to one side **102a** of protective shell **102**, while contactor **120** is optionally embedded in skirt **106** in a position spaced apart from male plug **116**. For example, contactor **120** is optionally embedded in skirt **106** spaced apart from male plug **116** and adjacent to one side **102b** of protective shell **102**.

[0110] Optionally, contactor body **128b** of adapter contactor **120** is recessed in outer surface **124** of unitary protective shell **102**. By example and without limitation, optional dam **132** is formed in outer surface **124** of unitary protective shell **102**, and adapter contactor **120** is further recessed within dam **132**. Electrical contacts **122** of contactor **120** are protected from damage by being recessed within dam **132** or otherwise within outer surface **124** of unitary protective shell **102**.

[0111] FIG. **19** and FIG. **20** illustrate individual adapter plug body **128a** of male plug **116** and contactor body **128b** of contactor **120** are interconnected in protective shell **102** by extended wires, traces or other electrical conductors **126**. For example, plurality of electrical conductors **126** are formed as elongated electrical cables, wires or other electrical conductors (three shown) that are extended for electrically interconnecting plurality of electrical connectors **118** of male adapter plug **116** with corresponding plurality of electrical contacts **122** of adapter contactor **120**. Electrical conductors **126** are fixedly positioned in protective shell **102**. For example, electrical conductors **126** are optionally embedded and overmolded in side skirt **106** of protective shell **102**.

Alternatively, electrical conductors **126** are optionally adhered to a surface of protective shell **102**, either an interior or an exterior surface.

[0112] Optionally, side **102a** and/or side **102b** of protective shell **102** are optionally adapted to receive plurality of electrical conductors **126** within skirt **106** of protective shell **102**. For example, portions of side **102a** and/or side **102b** of protective shell **102** are optionally thickened to accommodate additional volume of plurality of electrical conductors **126** within skirt **106**.

[0113] FIG. **21** illustrates converting adapter **114** configured as separated into a plurality of male plugs **116** (two shown) in a plurality of individual adapter plug bodies **128a** which are located in side skirt **106** adjacent to different sides **102a** in different areas of elastomer shell **102**. Each of the plurality of male plugs **116** is electrically coupled with electrical contacts **122** of contactor **120** in single contactor body **128b** mounted on side skirt **106** of protective shell **102**. Each individual adapter plug body **128a** of male plugs **116** and contactor body **128b** of contactor **120** are fixedly positioned in side skirt **106** of protective shell **102**. By example and without limitation, all male plugs **116** are optionally embedded in side skirt **106** adjacent to one or more different sides **102a** of protective shell **102**. For example, each individual adapter plug body **128a** is optionally overmolded into protective shell **102**. Contactor **120** is optionally embedded, for example overmolded, in side skirt **106** in a position spaced apart from one or more of male plugs **116**. For example, contactor **120** is optionally embedded in side skirt **106** spaced apart from male plug **116** and adjacent to one side **102b** of protective shell **102**.

[0114] As illustrated here, contactor body **128b** of adapter contactor **120** is recessed in outer surface **124** of side skirt **106** of unitary protective shell **102**. By example and without limitation, optional

dam **132** is formed in outer surface **124** of unitary protective shell **102**, and adapter contactor **120** is further recessed within dam **132**. Electrical contacts **122** of contactor **120** are protected from damage by being recessed within dam **132** or otherwise within outer surface **124** of unitary protective shell **102**.

[0115] As disclosed herein, dam **132** is a locator formed in outer surface **124** that cooperates with docking cradle **5** as a positioning interface for positively positioning unitary protective shell **102** relative to docking connector **3** for promoting mating therebetween.

[0116] As disclosed in each of FIG. **11** and FIG. **13**, positioning interface dam **132** operates as a locator on outer surface **124** of unitary protective shell **102** that cooperates with a socket receiver of docking cradle **5** for positively positioning contactor **120** with docking connector **3** for mechanical and electrical coupling therebetween. For example, dam **132** is projected or extended from outer surface **124** of unitary protective shell **102**, with adapter contactor **120** and electrical contacts **122** thereof being recessed within recess of locator dam **132**, as illustrated for example in FIG. **8**. As illustrated in each of FIG. **3**, FIG. **4** and FIG. **5**, docking connector **3** is recessed within a socket receiver that is recessed within interior surface of base receiver **9** of docking cradle tray **7**. Socket receiver thus cooperates with locator dam **132** for positively positioning unitary protective shell **102** relative to docking connector **3** in receiver **9** of docking cradle tray **7** for promoting mating therebetween, as illustrated herein.

[0117] FIG. **22** illustrates protective shell **102** having male plug **116** of converting adapter **114** separate and spaced apart from adapter contactor **120** thereof provided on an external surface **152** of flexible center panel **104** opposite of interior cavity **108**, as more clearly illustrated in subsequent figures.

[0118] FIG. **23** illustrates protective shell **102** having contactor **120** of converting adapter **114** provided on external surface **152** of flexible center panel **104** opposite of interior cavity **108**. Docking cradle **5** is alternatively configured to position docking connector **3** thereof in a position for coupling with contactor **120** on external surface **152** of flexible center panel **104** instead of being embedded in side skirt **106**.

[0119] FIG. **24** illustrates one embodiment of docking cradle **5**, wherein tray **7** is configured to lie substantially horizontally with base receiver **9** configured having an upper operational surface **156** and a lower interface surface **158** spaced apart by a peripheral side wall **160**. As illustrated here, tray **7** of docking cradle **5** may be structured with a generally round shape, but any generally square or rectangular, oval, kidney, or other regular or irregular shapes as are suitable and result in an equivalent structure.

[0120] Operational surface **156** of docking cradle tray **7** is formed with a female nest **157** formed of a generally flat or planar peripheral surface **164** forming a support rim surrounding a shallow socket receiver **166** recessed therein. Socket receiver **166** is, for example, of a generally round shape (shown) similarly to the outline shape of the docking cradle tray **7**, but may be of a generally square or rectangular, oval, kidney or other regular or irregular shape that optionally matches the shape of the outline shape of tray **7**. Recessed socket receiver **166** is formed with a floor **168** forming an electrical interface surface that is recessed relative to, and may be substantially parallel with, surrounding support rim **164**. An interior peripheral transition wall **170** provides a transition between socket floor **168** and surrounding rim support surface **164**. According to one embodiment of the invention, peripheral transition wall **170** is substantially perpendicular to the offset surfaces of the socket floor **168** and surrounding socket rim **164**. According to an alternative embodiment of the invention, peripheral transition wall **170** is optionally outwardly inclined between socket floor **168** and surrounding socket rim **164**, thereby providing an inclined transition between socket floor **168** and surrounding socket rim **164**. For example, an outwardly inclined peripheral transition wall **170** forms an angle in the range of about 30 degrees to 60 degrees from the plane of socket floor **168**. However, the angle is optionally in the much larger range of about 15 degrees or less to 75 degrees or more from the plane of socket floor **168**.

[0121] Docking cradle tray **7** optionally includes a mechanism for securing to an external mounting surface **S**. Accordingly, by example and without limitation, the docking cradle tray **7** is pierced by one or more fastener clearance holes **172** each sized to pass therethrough a mechanical fastener appropriate for attaching docking cradle tray **7** to an external mounting surface such as an intermediate mounting device, as discussed herein. Fastener clearance holes **172** optionally include coincidental annular depressions or recesses **174**, either countersinks or counter bores, for recessing the head of a threaded fastener passing therethrough below either of both of upper operational surface **156** and recessed socket floor **168**.

[0122] Alternatively, one or more fastener is optionally integrated with tray **7** of docking cradle **5**. For example, the heads of two pair of threaded studs are embedded in the molded structure of docking cradle tray **7** with their threaded shafts projecting out of lower interface surface **158** so as to be inserted through matching apertures in an external mounting surface and secured with nuts. In such instance fastener clearance holes **172** are filled with the fasteners' threaded shafts. Furthermore, recessed socket floor **168**, and surrounding socket rim **164** as well as peripheral transition wall **170** therebetween are optionally left completely unbroken, and docking cradle tray **7** thus presents a substantially solid and unbroken surface. Other mechanisms for attaching to external mounting surface **S**, such as adhesives or pressure sensitive adhesive tapes and films, are also contemplated and may be included or substituted without deviating from the scope and intent of the present invention.

[0123] Optionally, female nest **157** of docking cradle tray **7** is formed with one or more male rotational control features **176** that are configured to maintain a rotational orientation between unitary protective shell **102** of protective cover **100** and docking cradle tray **7**. By example and without limitation, male rotational control features **176** are configured as shallow intrusions into recessed socket receiver **166** by straight or curved (shown) portions of peripheral transition wall **170** between recessed socket floor **168** and surrounding socket rim **164**. Alternatively, peripheral transition wall **170** of socket receiver **166** is configured with any generally square or rectangular, oval, kidney, rectangular or other regular or irregular shapes, whereby rotational control features **176** are inherent in socket receiver **166** as provided by such non-round shapes.

[0124] According to one embodiment, shell **102** of protective cover **100** is retained in tray **7** of docking cradle **5** under the weight of portable electronic device **1** in shell **102**, i.e., by force of gravity. Positioning interface dam **132** of shell **102** cooperates with docking cradle **5** by nesting into socket receiver **166** recessed in docking cradle tray **7**.

[0125] Alternatively, one or more permanent magnets **178** are provided between shell **102** of protective cover **100** and tray **7** of docking cradle **5** for securing protective cover **100** in docking cradle **5**. By example and without limitation, one or more permanent magnets **178** are disposed within a magnet retention structure **180** that is adjacent to either recessed floor **168** of socket receiver **166**, or surrounding socket rim **164** (shown). When present, each permanent magnet **178** is, by example and without limitation, a thin round or disk-shape. Although the disk shape is convenient, other magnet shapes are equivalent and maybe substituted. Magnet **178** is optionally of the well-known rare-earth variety. Rare-earth magnets are very powerful in proportion to size and are therefore useful in practice of the present invention. However, other known and presently unknown magnets that are sufficiently powerful for practice of the invention are equivalent and may be substituted.

[0126] By example and without limitation, magnet retention structure **180** is embodied as a plurality of cavities that individually position each magnet **178** in close proximity to either recessed floor **168** of socket receiver **166**, or surrounding socket rim **164** of female nest **157**. According to one embodiment of the invention, each cavity of magnet retention structure **180** minimizes the distance between magnet **178** and socket's recessed floor **168**, or surrounding socket rim **164**. Attenuation of magnetic fields generated by magnets **178** is thereby minimized, and the magnetic field generated remains sufficiently powerful to ensure retention of most handheld electronic and

other small devices by magnetic attraction to cooperating docking cradle 5.

[0127] According to one embodiment, cavities of magnet retention structure **180** are sized to match magnets **178**, either as a slip fit or a compression fit such that each magnet **178** is inserted under at least light pressure and as much as a press fit.

[0128] Magnet retention cavities **180** may be provided on lower interface surface **158** of tray **7** of docking cradle **5** opposite from operational surface **156**. Magnets **178** in retention cavities **180** are retained in close proximity of recessed floor **168** of socket receiver **166**, or surrounding socket rim **164** of female nest **157**. When docking cradle tray **7** is formed of a substantially rigid plastic or other elastomeric material having a property of being resiliently responsive to slight deformations, and magnet retention cavity **180** is optionally sized as a compression fit for magnet **178**, whereby magnet **178** is retained by radial compression of undersized cavity **180** which is resiliently deformed by magnet **178** upon insertion. Therefore, other retention elements are avoided such as bi-adhesive tape as taught by both Tarulli in U.S. Pat. No. 5,992,807 and Won in U.S. Pat. No. 6,149,116 or the bayonets in mating apertures as taught by Rielo in U.S. Pat. No. 5,895,018 entitled, MAGNETIC SUPPORT ATTACHMENT, issued Apr. 20, 1999, which is incorporated herein by reference.

[0129] As disclosed herein, docking connector **3** includes plurality of biasing electrical contacts **11** configured, for example, as springloaded pogo pins that are electrically coupled to electrical leads **13** in wire or cable **15**. Typically, docking connector **3** includes a minimum number of at least two biasing electrical contacts **11**, but is not limited to having greater numbers of contacts **11**.

[0130] FIG. 25 illustrates unitary protective shell **102** of protective cover **100** formed with locator dam **132** of shell **102** as a complementary male nesting appendage **159** configured to seat in female nest **157**. For example, locator dam **132** of complementary male nesting feature **159** is configured for mating with recessed socket receiver **166** of horizontal base receiver **9** in tray **7** of docking cradle **5** as illustrated in FIG. 24. Complementary locator dam **132** is projected or extended from external surface **152** of flexible center panel **104** opposite of interior cavity **108** where portable electronic device **1** is seated. Complementary locator dam **132** is configured with a generally round shape when recessed socket receiver **166** is so configured. Alternatively, locator dam **132** is configured with any generally square or rectangular, oval, kidney, rectangular or other regular or irregular shape when recessed socket receiver **166** of female nest **157** in operational surface **156** of docking cradle tray **7** is so configured. Thus, complementary locator dam **132** is a shallow projection formed with a substantially planar contactor surface **182** offset from external surface **152** of flexible center panel **104** to distance about the depth of recessed floor **168** in socket receiver **166** of docking cradle tray **7**. Thus, external surface **152** of flexible center panel **104** of protective shell **102** forms a lower rim of male nesting appendage **159** that engages peripheral support rim **164** of docking cradle tray **7** when complementary locator dam **132** nests in recessed socket receiver **166**.

[0131] When peripheral transition wall **170** of female socket receiver **166** is inclined between socket floor **168** and surrounding socket rim **164**, complementary male locator dam **132** optionally includes a exterior peripheral transition wall **184** extended between contactor surface **182** and external surface **152** of flexible center panel **104** and formed with a substantially matching incline. Accordingly, inclined complementary male locator dam **132** nests in incline of recessed socket receiver **166** of female nest **157**.

[0132] When socket receiver **166** of female nest **157** is configured with one or more male rotational control features **176**, complementary male locator dam **132** is configured with one or more matching rotational control features **186**. For example, when rotational control features **176** of female socket receiver **166** are configured as male peripheral protrusions of transition wall **170** into interior of socket **166** between recessed socket floor **168** and surrounding socket rim **164**, rotational control features **186** are configured as matching female indents or intrusions into periphery of complementary locator dam **132** that are positioned and sized to mate with male rotational control features **176** in socket transition wall **170**.

[0133] Alternatively, matching male and female rotational control features **176** and **186** are optionally reversed between female socket receiver **166** and complementary locator dam **132** such that female rotational control features **176** are presented in complementary locator dam **132** of shell **102** as male peripheral protrusions of transition wall **184**, and matching female rotational control features **186** are presented as complementary female indents or intrusions into periphery of transition wall **170** into rim **164** surrounding female socket receiver **166**.

[0134] Thus, as disclosed here, male positioning interface **132** includes a combination of both offset projection contactor surface **182** with recessed female rotational control features **186** formed in its periphery and which are, for example, coplanar with surrounding peripheral rim formed by external surface **152** of flexible center panel which contacts rim **164** surrounding socket receiver **166** when seated in female nest **157** of docking cradle tray **7**. Recessed female rotational control features **186** of male positioning interface **132** engage male rotational control features **176** of docking cradle tray **7** in female nest **157**, whereby shell **102** is rotationally oriented relative to base receiver **9** of tray **7**. Accordingly, complementary locator dam **132** on external surface **152** of protective shell **102** nests with female socket receiver **166** in base receiver **9** of docking cradle tray **7**, while mating male and female rotational control features **176**, **186** control rotation of protective shell **102** relative to base receiver **9** of tray **7**.

[0135] When tray **7** of docking cradle **5** includes optional magnets **178** adjacent to recessed floor **168** or surrounding rim **164** of socket receiver **166**, as shown in FIG. **24**, center panel **104** of protective shell **102** is provided with a magnetically permeable member **188** embodied as a thin sheet or plate of a ferrous or other highly magnetically permeable ferromagnetic material, such as iron, nickel, cobalt or another ferromagnetic material or alloy. For example, magnetically permeable member **188** is embedded in center panel **104** of protective shell **102** in a complementary position for attraction by magnets **178**. Magnets **178** and magnetically attractive member **188** thus form a magnetic attraction couple operable between shell **102** of protective cover **100** and tray **7** of docking cradle **5** for securing protective cover **100** in docking cradle **5**.

[0136] Contactor **120** of converting adapter **114** provided on external surface **152** of flexible center panel **104** is suitably configured to electrically couple with plurality of biasing electrical contacts **11** configured as springloaded pogo pins in recessed socket receiver **166** of horizontal tray **7**. For example, plurality of electrical contacts **122** of contactor **120** are configured as a plurality of concentric contact rings of an electrically conductive material, such as copper, silver, gold, platinum, or another electrically conductive material. Contact rings **122** of contactor **120** are coupled to electrical conductors **126** for electrically interconnecting with plurality of electrical connectors **118** of male adapter plug **116**, in either sequentially or non-sequentially ordered pairs, as disclosed herein. Contact rings **122** of contactor **120** are positioned to mate with different pogo pin contacts **11** in recessed socket receiver **166** of base receiver **9** in docking cradle tray **7**.

[0137] When electrical contacts **122** of contactor **120** are configured as the plurality of concentric contact rings, cooperating male and female rotational control features **176** and **186** are optionally eliminated because contact rings **122** normally mate with different pogo pin contacts **11** in any orientation of shell **102** relative to docking cradle tray **7** regardless of rotation (FIG. **25**, arrows **190**) of positioning interface dam **132** in female socket receiver **166**. However, cooperating male and female rotational control features **176** and **186** may be present for providing relative rotational relationship between protective cover **100** and docking cradle **3**, if desired.

[0138] FIG. **26** is a cross-section view that illustrates elastomeric shell **102** of protective cover, or skin, **100** seated in tray **7** of docking cradle **3**. As illustrated, complementary male nesting appendage **159** of protective cover **100** is nested with female nest **157** of docking cradle **5**, whereby shell **102** is retained in docking cradle tray **7**. For example, shell **102** is secured by the weight of portable electronic device **1** in shell **102**, i.e., by force of gravity, as disclosed herein. Shell **102** cooperates with docking cradle **5** by nesting positioning interface dam **132** into socket receiver **166** recessed in docking cradle tray **7**. For example, positioning interface dam **132** is rotated (FIG. **25**,

arrows **190**) until optional female rotational control features **186**, if present, mate with matching male rotational control features **176**, if present, of tray **7** of docking cradle **5**. Thereafter, peripheral transition wall **184** of cooperates with matching transition wall **170** to position locator dam **132** in socket receiver **166** as shell **102** approaches docking cradle tray **7**. When positioning interface dam **132** of complementary male nesting feature **159** is fully nested in socket receiver **166** of female nest **157**, external surface **152** of flexible center panel **104** contacts peripheral support rim **164** of female socket receiver **166**, and contactor surface **182** of elastomer shell **102** is juxtaposed with recessed socket floor **168**. Engagement of cooperating peripheral transition walls **184** and **170** of projected positioning interface dam **132** and recessed socket receiver **166** operate to guide electrical contacts **122** of contactor **120** into engagement with biasing electrical contacts **11** of base receiver **9** of docking cradle **3**.

[0139] According to one embodiment, electrical contacts **122** of contactor **120** engage biasing electrical contacts **11** of base receiver **9** of docking cradle **3** under the weight of shell **102** with electronic device **1** therein, i.e., by force of gravity, as disclosed herein.

[0140] Alternatively, when optional magnets **178** are present, magnetic attraction between magnets **178** and magnetically permeable member **188** operates to secure shell **102** of protective cover **100** in nested engagement with tray **7** of docking cradle **5** and engage electrical contacts **122** of contactor **120** with biasing electrical contacts **11** of base receiver **9** of docking cradle **5**.

[0141] When protective cover, or skin, **100** is retained in tray **7** of docking cradle **3** under force of gravity, simply lifting shell **102** will release protective cover **100** for removal from cradle **3**.

[0142] Alternatively, when optional magnets **178** are present, magnetic attraction between magnets **178** and magnetically permeable member **188** operates to secure shell **102** of protective cover **100** in nested engagement with tray **7** of docking cradle **5**. However, simply rotating shell **102** (FIG. **25**, arrows **190**) causes positioning interface dam **132** to be rotated until female rotational control features **186** interfere with matching male rotational control features **176** of tray **7** of docking cradle **5**. Such interference causes shell **102** to separate from base receiver **9** and attenuate the magnetic attraction, whereby protective cover **100** is released for removal from cradle **3**.

[0143] FIG. **27**, FIG. **28** and FIG. **29** together illustrate an alternative embodiment of cover or skin **100** and docking cradle **5**. As disclosed herein, the relative locations of magnets **178** and magnetically permeable member **188** are inverted. For example, one or more magnets **178** are disposed within a magnet retention structure **180** adjacent to external surface **152** of flexible center panel **104** instead of being in raised rim **164** surrounding socket receiver **166** of female nest **157** in docking cradle tray **7**. As illustrated by example and without limitation, one or more magnetically permeable members **188** each embodied as a thin sheet or plate of a ferrous or other highly magnetically permeable ferromagnetic material, such as iron, nickel, cobalt or another ferromagnetic material or alloy, as disclosed herein, are embedded in either recessed floor **168** of female socket receiver **166**, or surrounding socket rim **164** (shown).

[0144] Additionally, one or more permanent magnets **178** mounted in complementary male nesting appendage **159** of protective cover **100**. For example, magnets **178** are embedded in center panel **104** of protective shell **102** for cooperating with magnetically permeable members **188** for magnetically securing protective cover **100** in docking cradle **5**.

[0145] FIG. **30**, FIG. **31** and FIG. **32** illustrate an alternative embodiment of cooperating device cover **100** and docking cradle **5**, wherein the positioning interface feature provided between contactor **120** of converting adapter **114** and docking connector **3** formed by male nesting appendage **159** and mating female nest **157** of docking cradle tray **7** is inverted. The inverted embodiment of the positioning interface feature of female nest **157** of and complementary male nesting feature **159** is formed by locator dam **132** being configured as the female receiver which is provided on external surface **152** of center panel **104** of shell **102**, while base receiver **9** of docking cradle tray **7** is configured as the cooperating male locator.

[0146] FIG. **30** illustrates female nest **157** being provided on protective shell **102** by locator dam

132 which is configured here with shallow socket receiver **166** from docking cradle tray **7**. Socket receiver **166** of locator dam **132** is surrounded by generally flat or planar peripheral surface **164** forming a support rim thereabout that is projected from external surface **152** of flexible center panel **104** opposite of interior cavity **108** of shell **102**. Socket receiver **166** of dam **132** includes interior peripheral transition wall **170** that provides transition between socket floor **168** and surrounding rim support surface **164**. Transition wall **170** is formed with rotational control features **176** that are configured to positively rotationally orient elastomer shell **102** of protective cover **100** relative to docking cradle tray **7**, as disclosed herein.

[0147] Thus, as disclosed here, male positioning interface **132** includes a combination of both contactor surface recessed in floor **168** of socket receiver **166** with projected rotational control features **176** formed in its periphery and which are, for example, coplanar with surrounding peripheral contact rim **164** surrounding socket floor **168**. Contact rim **164** engages operational surface **156** of docking cradle tray **7** when female nest **157** is seated over complementary male nesting appendage **159** of docking cradle tray **7**. Projected rotational control features **176** of female positioning interface **132** engage recessed rotational control features **186** of docking cradle tray **7** in male nesting appendage **159**, whereby shell **102** is rotationally oriented relative to base receiver **9** of tray **7**.

[0148] Optionally, shell **102** of protective cover **100** is retained in tray **7** of docking cradle **5** under weight, i.e., by force of gravity, as disclosed herein. Alternatively, optional magnets **178** and magnetically permeable member **188**, when present, may be provided in either of docking cradle tray **7** or protective shell **102**, as disclosed herein. As illustrated here by example and without limitation, magnets **178** are provided in magnet retention structure **180** that is adjacent to either recessed floor **168** of socket receiver **166**, or surrounding socket rim **164** (shown).

[0149] Contactor **120** of converting adapter **114** may be configured as a plurality of concentric contact rings **122** of an electrically conductive material, as disclosed herein. Contactor **120** is positioned on floor **168** of socket receiver **166**.

[0150] FIG. **31** illustrates tray **7** of docking cradle **5** being configured to lie substantially horizontally with base receiver **9** configured having its upper operational surface **156** formed of a generally flat plane. Operational surface **156** of tray **7** is configured with complementary male nesting appendage **159** from protective cover **100**.

[0151] Substantially planar contactor surface **182** is offset or projected from upper operational surface **156** of docking cradle tray **7** and is configured with any generally round, square or rectangular, oval, kidney, rectangular or other regular or irregular shape and sized to fit snugly within recessed socket receiver **166** of female locator dam **132** of nest **157** on protective shell **102**. Thus, upper operational surface **156** of docking cradle tray **7** engages peripheral support rim **164** of locator dam **132** in protective shell **102** when projected contactor surface **182** of complementary male nesting appendage **159** nests in recessed socket receiver **166** in female locator dam **132** of nest **157**.

[0152] Thus, as disclosed here, complementary male nesting appendage **159** of docking cradle tray **7** includes a combination of both contactor surface **182** projected from upper operational surface **156** of base receiver **9** with female rotational control features **186** recessed in its periphery and which are, for example, coplanar with surrounding peripheral contact rim formed as operational surface **156**. Operational surface **156** engages contact rim **164** of peripheral surface **164** forming a support rim surrounding socket receiver **166** of locator dam **132** that is projected from external surface **152** of flexible center panel **104** of shell **102** (shown in FIG. **30**) when female nest **157** is seated over complementary male nesting appendage **159** of docking cradle tray **7**. Projected male rotational control features **176** of female positioning interface **132** engage recessed female rotational control features **186** of docking cradle tray **7** in male nesting appendage **159**, whereby shell **102** is rotationally oriented relative to base receiver **9** of tray **7**.

[0153] As disclosed herein, docking connector **3** includes plurality of biasing electrical contacts **11**

configured, for example, as springloaded pogo pins that are electrically coupled to electrical leads **13** in wire or cable **15**.

[0154] Docking cradle tray **7** is optionally pierced by fastener clearance holes **172** for attaching docking cradle tray **7** to an external mounting surface, as discussed herein. Fastener clearance holes **172** optionally include coincidental annular depressions or recesses **174**, either countersinks or counter bores, as discussed herein, for recessing the head of a threaded fastener passing therethrough below projected contactor surface **182** of complementary male nesting appendage **159**. As further disclosed herein, other mechanisms for attaching to external mounting surface **S**, such as adhesives or pressure sensitive adhesive tapes and films, are also contemplated and may be included or substituted without deviating from the scope and intent of the present invention.

[0155] Exterior peripheral transition wall **184** of complementary projected male nesting appendage **159** is optionally inclined for cooperatively nesting in incline of recessed socket receiver **166** for positively positioning unitary protective shell **102** relative to docking connector **3** in receiver **9** of docking cradle tray **7**.

[0156] When locator dam **132** of protective shell **102** is formed with male rotational control features **176**, tray **7** of docking cradle **5** may be formed with matching female rotational control features **186** that mate with male rotational control features **176** and cooperate for controlling rotational orientation between protective shell **102** and docking cradle tray **7**.

[0157] Alternatively, matching male and female rotational control features **176** and **186** are optionally reversed between female socket receiver **166** of locator dam **132**, forming female nest **157** on protective shell **102**, and complementary male nesting appendage **159** of docking cradle tray **7** such that rotational control features **186** are presented as female indents or intrusions into periphery of transition wall **170** into rim **164** surrounding female socket receiver **166** of locator dam **132**, and matching rotational control features **176** are presented as complementary male peripheral protrusions of transition wall **184**.

[0158] FIG. **32** is a cross-section view, wherein female nest **157** is provided on protective shell **102** by locator dam **132**, and complementary male nesting appendage **159** is provided on tray **7** of docking cradle **5** by base receiver **9**. Accordingly, positioning interface between shell **102** of protective cover **100** and tray **7** of docking cradle **5** is provided by female nest **157** on protective shell **102** being received onto complementary male nesting appendage **159** of base receiver **9**.

[0159] When protective cover **100** is secured by the weight of the portable electronic device **1** in shell **102**, i.e., by force of gravity, as disclosed herein, tray **7** of docking cradle **5** is expected to be presented with complementary male nesting appendage **159** of base receiver **9** oriented generally upright for receiving female nest **157** of protective shell **102** thereonto. Alternatively, one or more permanent magnets **178** and magnetically permeable members **188** are disposed in complementary locations for magnetic attraction between shell **102** of protective cover **100** and tray **7** of docking cradle **5**. For example, when optional magnets **178** and magnetically permeable member **188** are present, as illustrated here, rotation of docking cradle tray **7** is possible with electronic device **1** in protective cover **100** being releasably magnetically mounted thereon. Depending upon the strength of magnetic attraction as a function of selected magnets **178** and magnetically permeable member **188**, tray **7** of docking cradle **5** can be tilted or rotated to a greater extreme, even to inverted with docking cradle tray **7** being upside-down, with electronic device **1** secured therein by magnetic attraction between protective cover **100** and docking cradle tray **7** sufficiently that electrical contacts **122** of shell contactor **120** remain mechanically and electrically coupled to electrical contacts **11** of docking connector **3** sufficiently for operation of electronic device **1**.

[0160] As illustrated by example and without limitation, one or more magnetically permeable members **188** each embodied as a thin sheet or plate of a ferrous or other highly magnetically permeable ferromagnetic material, such as iron, nickel, cobalt or another ferromagnetic material or alloy, as disclosed herein, are embedded in either recessed floor **168** of female socket receiver **166**, or surrounding socket rim **164** (shown) of locator dam **132** on protective shell **102**. Additionally,

one or more permanent magnets **178** mounted in complementary male nesting appendage **159** of protective cover **100**. For example, magnets **178** are embedded in center panel **104** of protective shell **102** for cooperating with magnetically permeable members **188** for magnetically securing protective cover **100** in docking cradle **5**.

[0161] Alternatively, relative locations of magnets **178** and magnetically permeable member **188** are optionally inverted, as disclosed herein, wherein one or more magnets **178** are disposed within a magnet retention structure **180** in female nest **157** of locator dam **132** on protective shell **102**, and one or more magnetically permeable members **188** are disposed in a complementary location in operational surface **156** of tray **7** adjacent to male nesting appendage **159** of base receiver **9**.

[0162] As disclosed herein, interior transition wall **170** of socket receiver **166** cooperates with exterior peripheral transition wall **184** of complementary projected male nesting appendage **159** for positively positioning unitary protective shell **102** relative to docking connector **3** in receiver **9** of docking cradle tray **7**.

[0163] Furthermore, when locator dam **132** of protective shell **102** is formed with rotational control features **176**, tray **7** of docking cradle **5** may be formed with complementary rotational control features **186** that mate with rotational control features **176** and cooperate therewith for controlling rotational orientation between protective shell **102** and docking cradle tray **7**.

[0164] FIG. **33** illustrates protective cover **100**, wherein complementary male nesting **15** appendage **159** is configured as positioning interface dam **132** in side skirt **106** of protective shell **102**, as similarly shown in FIG. **6**, for being received into female nest **157** of base receiver **9**, as similarly shown in FIG. **11**. In contrast, here positioning interface dam **132**, which includes contactor **120** and its contacts **122**, is spaced away from male plug **116** which is extended into cavity **108** of shell **102** at a position in a different side skirt **106** of protective shell **102** away from contactor **120** and its contacts **122**, as disclosed herein.

[0165] Positioning interface dam **132** of male nesting appendage **159** is optionally formed with magnet retention structures **180** adjacent to either side of contactor **120** of converting adapter **114**, whereby magnets **178** are mounted adjacent to contactor **120**. Accordingly, protective cover **100** is configured to cooperate with tray **7** of docking cradle **5** having one or more magnetically permeable members **188**, such that magnetic attraction of protective shell **102** to tray **7** operates to secure protective cover **100** in docking cradle **5** when male nesting appendage **159** is seated in complementary female nest **157**, as disclosed herein.

[0166] FIG. **34** illustrates protective cover **100** having electronic device **1** installed therein. Protective cover **100** here is similar to the embodiment illustrated in FIG. **33**, except here positioning interface dam **132** of male nesting appendage **159** instead includes magnetically permeable member **188** embodied as one or more thin sheets or plates of a ferrous or other highly magnetically permeable ferromagnetic material, such as iron, nickel, cobalt or another ferromagnetic material or alloy. For example, magnetically permeable member **188** is embedded in locator dam **132** on outer surface **124** of protective shell **102** in a complementary position for cooperating with one or more magnets **178** positioned in base receiver **9** of docking cradle tray **7**. Thus, when male nesting appendage **159** is seated in complementary female nest **157**, magnets **178** and magnetically permeable members **188** cooperate for securing protective cover **100** with electronic device **1** in docking cradle **5**, as disclosed herein.

[0167] FIG. **35** is a cross-section taken through protective cover **100** of FIG. **33**, wherein male plug **116** and contactor **120** of electrical adapter **114** are spaced apart in different areas of side skirt **106** of protective shell **102**. As illustrated here, electrical contacts **122** of contactor **120** embedded in body **128** of adapter **114** are electrically coupled to one or more of electrical connectors **118** of plug **116** via electrical cables, wires, traces or other electrical conductors **126** (one shown for clarity). Portable electronic device **1** is received within cavity **108** of elastomer shell **102** with female input/output socket **27** engaged with male plug **116** of adapter **114** in skirt **106** adjacent to one side **102a** of protective shell **102**.

[0168] Positioning interface dam **132** is formed in outer surface **124** of unitary protective shell **102**, and adapter contactor **120** is further recessed within locator dam **132** to form complementary male nesting appendage **159** for seating in female nest **157**. Here, locator dam **132** includes magnets **178** disposed within magnet retention structure **180** for magnetic attraction of magnetically permeable members **188** in base receiver **9** of docking cradle tray **7**.

[0169] Alternatively, magnets **178** are in base receiver **9** of docking cradle tray **7**, and cooperating magnetically permeable members **188** are flanking electrical contacts **122** of contactor **120** in locator dam **132** exterior of protective shell **102**.

[0170] FIG. **36** is a cross-section taken through protective cover **100** of FIG. **34**, wherein male plug **116** and contactor **120** of electrical adapter **114** are spaced apart in different areas of side skirt **106** of protective shell **102**. As illustrated here, electrical contacts **122** of contactor **120** embedded in body **128** of adapter **114** are electrically coupled to one or more of electrical connectors **118** of plug **116** via electrical cables, wires, traces or other electrical conductors **126** (one shown for clarity). Portable electronic device **1** is received within cavity **108** of elastomer shell **102** with female input/output socket **27** engaged with male plug **116** of adapter **114** in skirt **106** adjacent to one side **102a** of protective shell **102**.

[0171] Positioning interface dam **132** is formed in outer surface **124** of unitary protective shell **102**, and adapter contactor **120** is further recessed within locator dam **132** to form complementary male nesting appendage **159** for seating in female nest **157**. Here, locator dam **132** alternatively includes magnetically permeable ferromagnetic member **188**, as disclosed herein. For example, ferromagnetic member **188** is embedded in locator dam **132** in side skirt **106** of protective shell **102** in a complementary position for attraction by magnets **178** in base receiver **9** of docking cradle tray **7**, as disclosed herein.

[0172] FIG. **37** illustrates female nest **157** of base receiver **9** in docking cradle **5**, including one or more ferromagnetic members **188**. For example, ferromagnetic members **188** are exposed in positions flanking biasing electrical contacts **11** of docking connector **3**. Accordingly, when positioning interface dam **132** of protective cover **100** in side skirt **106** of shell **102** is optionally provided with magnets **178** mounted adjacent to either side of contactor **120**, as illustrated for example in FIG. **33**, ferromagnetic members **188** are positioned to cooperate with magnets **178** such that magnetic attraction of protective shell **102** to tray **7** operates to secure protective cover **100** in docking cradle **5** when male nesting appendage **159** is seated in complementary female nest **157**, as disclosed herein. Docking connector **3** may also include optional guide pins **17**, as disclosed herein, that are adapted for mating with optional mating guide pin receivers **130** if present in contactor **120** of electrical adapter **114**.

[0173] FIG. **38** illustrates female nest **157** of alternative base receiver **9**, wherein one or more magnets **178** are substituted for ferromagnetic members **188** shown in FIG. **37**. For example, magnets **178** are positioned flanking biasing electrical contacts **11** of docking connector **3**. Accordingly, when positioning interface dam **132** of protective cover **100** in side skirt **106** of shell **102** is optionally provided with ferromagnetic members **188** mounted adjacent to either side of contactor **120**, as illustrated for example in FIG. **34**, magnets **178** are positioned to cooperate with ferromagnetic members **188** such that magnetic attraction of protective shell **102** to tray **7** operates to secure protective cover **100** in docking cradle **5** when male nesting appendage **159** is seated in complementary female nest **157**, as disclosed herein. Docking connector **3** may also include optional guide pins **17**, as disclosed herein, that are adapted for mating with optional mating guide pin receivers **130** if present in contactor **120** of electrical adapter **114**.

[0174] FIG. **39** illustrates another arrangement of electrical contacts **122** for the cover **100**. In this arrangement, there are ten electrical contacts **122** arranged in two rows. In the illustrated embodiment, the rows are staggered, but it will be recognized that the rows can be aligned with each other, instead of staggered. The docking cradle **5** can have the same number and arrangement of electrical contacts **11**, although in some embodiments, the docking cradle may have fewer or

more contacts than the cover **100**.

[0175] In at least some embodiments, different covers **100** can be provided for different types of portable electronic devices **1**, but the arrangement of electrical contacts **122** is the same so that a variety of different portable electronic devices **1** (including devices from different manufacturers) can be coupled to the same docking cradle **5** using an appropriate cover **100**. It will be understood that the internal arrangement of electrical conductors **126** (which connect the electrical contacts **122** of the adapter **114** to the electrical connectors **118** of the male plug **116**) of the different covers **100** may be different depending on the type of portable electronic device **1** that fits in the cover **100**. Moreover, in at least some embodiments, one or more of the electrical contacts **122** may not be coupled to a corresponding electrical contact **118** of the adapter **114**, particularly if there is not a corresponding contact in the portable electrical device **1** for which the cover **100** is designed.

[0176] FIG. **41** illustrates an arrangement of twelve electrical contacts **122** in two staggered rows. It will be understood that the cover **100** can include any number of electrical contacts **122** including, but not limited to, two, three, four, five, six, seven, eight, nine, ten, twelve, fourteen, sixteen, eighteen, twenty, or more electrical contacts. The electrical contacts **122** can be arranged in any regular or irregular pattern. For example, three electrical contacts can be arranged in a straight line, an equilateral triangle, an isosceles triangle, a right triangle, or any other suitable arrangement. As another example, the electrical contacts **122** can be arranged in a single line or in two or more rows (e.g., two, three, four, or more rows) which may be aligned or staggered relative to each other and which may have the same or different numbers of contacts in each row. As yet another example, some or all of the electrical contacts **122** can be arranged at vertices of a regular or irregular polygon. For example, five electrical contacts **122** can be arranged at vertices of a pentagon, six electrical contacts **122** can be arranged at vertices of a hexagon, or eight electrical contacts can be arranged at vertices of an octagon.

[0177] FIGS. **39** and **41** also illustrate a dam **132** which has an asymmetric shape that can facilitate correctly coupling the cover **100** to the docking cradle **5**. The dam **132** defines an asymmetric positioning interface. In this case, the dam **132** has a first edge **191** that is straight and a second edge **192**, opposite the first edge, that is curved. The female nest **157** of the docking cradle **5** can be shaped similarly to the dam **132** so that the dam of the cover **100** only fits easily into the female nest of the docking cradle in one orientation. This asymmetric positioning interface of the cover facilitates correct coupling of the cover with the docking cradle **5** so that the electrical contacts **122** correctly and reliably mate with corresponding contacts on the docking cradle.

[0178] FIG. **40** illustrates a portion of the adapter **114** and includes the male plug **116**, electrical contacts **122**, magnets **178** (or ferromagnetic members **188**), and body **128**. The adapter **114** is positioned within the cover **100** with the side skirt **116** and dam **132** formed around portions of the adapter. In at least some embodiments, a portion of the shell **102** or dam **132** also covers a face **193** of the body **128** and the magnets **178** leaving a portion of the electrical contacts **122** exposed, as illustrated in FIGS. **39** and **41**.

[0179] A cover for a portable electronic device can also include an additional pocket for a peripheral device. FIGS. **42-46** illustrate an embodiment of a cover **100** that can receive a portable electronic device **1** and a peripheral device **50**. Examples of peripheral devices include, but are not limited to, a printer, a scanner, a card reader, a magnetic strip reader, a RFID reader, a NFC reader, a speaker, a battery, a camera, a light, a keyboard, a human interface device (e.g., a mouse, trackball, or the like), a medical device (e.g., a thermometer, glucose sensor, blood pressure monitor, imager, or any other suitable medical device), or any other suitable peripheral device. Other examples of peripheral devices can include illumination devices that can provide LED lights, black lights, infrared lights, lasers, or any other lighting devices, or any combination thereof, and can be used to provide general illumination, night vision, video or photography illumination or enhancement, signaling, pointing, measuring, scanning, alignment, or any other feature or use or any combination thereof. Such illumination devices may be designed to be used with any device

within the cover such as, for example, a cell phone or smart phone, a terminal, a computer, a testing device, a tablet, or the like, or any combination thereof. Other examples of peripheral devices can be tool devices such as, for example, sensors (for detecting, for example, temperature, pressure, leveling, angles, altitude, light, density, x-rays, magnetic fields, acceleration, speed, velocity, distance, energy, current or energy flow, force, durometer, mass/weight, torque, resistance, capacitance, voltage, reactance, or the like or any combination thereof) or a fingerprint reader.

[0180] The cover **100** may be designed for a specific type of peripheral device or may accommodate multiple types of peripheral devices. In addition, although the illustrated embodiment permits inclusion of a single peripheral device in the cover, it will be understood that other embodiments can permit inclusion of multiple (e.g., two, three, four, or more) peripheral devices in one or more peripheral pockets in the cover.

[0181] The cover **100** includes a shell **102** with a center panel **104**, a side skirt **106**, a lip **110**, a mouth opening **112**, and an interior cavity **108** to receive the portable electronic device **1**. In addition, the cover **100** includes a peripheral cavity **208** that forms a peripheral pocket for receiving the peripheral device **50**. The peripheral cavity **208** can be formed using the center panel **104** and optionally part of the side skirt **106** or any other portion of the cover **100**. The peripheral cavity **208** defines a device surface **205** and one or more side surfaces **207**. In the illustrated embodiment, the peripheral cavity **208** and interior cavity **108** are contiguous and the peripheral device **50** can be inserted into the peripheral cavity **208** through the mouth opening **112** and interior cavity **108**, as illustrated in FIG. **44**. The portable electronic device **1** can then be inserted into the interior cavity **110** through the mouth opening, as illustrated in FIG. **45**. In other embodiments, the cover **100** may include a peripheral opening (not shown) through which the peripheral may be inserted into the peripheral cavity **208**. In such embodiments, the peripheral cavity **208** may be contiguous with the interior cavity **108** or the peripheral cavity and interior cavity can be separated by a wall of the cover **100**.

[0182] In addition to the male plug **116**, the cover **100** includes a peripheral plug **216**, as illustrated in FIG. **43**. The peripheral plug **216** fits into the input/output socket of the peripheral device **50**. The peripheral plug **216** may be designed for a particular peripheral device or may be suitable for multiple different devices. For example, the peripheral plug **216** (or the male plug **116** or both) can be a USB or microUSB plug designed to fit into a USB or microUSB port, respectively, in the peripheral device **50**.

[0183] The peripheral plug **216** includes one or more electrical connectors **218** that connect to contacts within the peripheral device **50**. The one or more electrical connectors **218** are coupled to the electrical contacts **122** of the adapter **114** of the cover **100** using conductors (not shown) that extend through the cover. This arrangement can be used to provide power to the peripheral device **50** via the adapter **114** and one or more electrical connectors **218**. In at least some embodiments, the adapter **114** and associated conductors can be arranged so that the peripheral device **50** provides power, data, or both to the portable electronic device **1** through the adapter. For example, the adapter **114** may couple an electrical connector **218** of the peripheral plug **216** and an electrical connector **118** of the male plug **116** to the same electrical contact **122** of the adapter so that the peripheral device **50** can provide power or data to the portable electronic device **1** through the adapter. It will be understood that there are other mechanisms for sharing data between the peripheral device **50** and the portable electronic device **1** such as Bluetooth, NFC, and the like.

[0184] The cover **100** can include one or more openings **194** through the cover into the peripheral cavity **218**. These openings **194** can allow access to components of the peripheral device **50** such as, for example, a jack, a scanner, a printer, a sensor, or the like. Similarly, the cover **100** can include one or more openings **196** into the interior cavity **118** to allow access to components of the portable electronic device. For example, the openings **196** can allow access to a camera, a headphone jack, a switch, a speaker, a microphone, or the like of the portable electronic device **1**. In at least some embodiments, the placement, number, and size of the openings **196** of the cover are

specifically arranged based on the type of portable electronic device for which the cover **100** is designed. In some embodiments, the cover **100** can include soft buttons **198** formed in the cover (for example, in the side skirt **106**) at positions that can coincide with buttons on the portable electronic device **1**. The soft buttons **196** can be, for example, part of the material of the cover with a surrounding indentation to identify placement of the soft button and facilitate actuation of the soft button. The soft buttons **196** are configured and arranged so that actuation of the soft button also actuates the underlying button of the portable electronic device **1**. Such soft buttons **196** may coincide with power, volume, and other buttons on the portable electronic device **1**. Any of the covers described herein can include one or more openings **196**, one or more soft buttons **198**, or any combination thereof.

[0185] Any of the covers **100** described herein can also include a tag element **199** disposed within or on the cover, as illustrated in FIG. **43**. The tag element **199** can be, for example, a RFID tag or a NFC tag that can be used to identify the cover or device disposed within the cover. The tag element **199** can be molded into the cover **100** or may be disposed on an interior or exterior surface of the cover. In at least some embodiments, the tag element **199** can be associated with a fob **197** or other tag query device that a user can carry and which can be used to wirelessly query the tag element **199** to determine if the cover **100** is within a range of the fob or other tag query device. The fob **197** can be activated by the user to assist in finding the cover or associated portable electronic device. Alternatively or additionally, the fob **197** may include an audible alarm, vibratory alarm, or both that can be automatically activated when the fob exceeds a threshold distance from the cover **100** to warn the user that the cover **100** and associated portable electronic device are left behind (for example, left on the docking cradle). In some embodiments, the user may select whether the fob **197** provides an audible alarm, a vibratory alarm, or both. In some embodiments, the user may program the threshold distance.

[0186] FIGS. **47-49** illustrate additional embodiments of the docking cradle **5**. These docking cradles **5** include a tray **7** (not shown in FIG. **49**) and a base receiver **9** upon which a portable electronic device **1** within a cover **100** can be placed. The docking cradle **5** of FIG. **47** includes a clamp **19** and an arm **21**. The docking cradle **5** of FIG. **48** is a stand-alone cradle that can be placed on any stable surface, such as a desk, shelf, credenza, table, or the like. The docking cradle **5** of FIG. **49** can be mounted on a wall or other surface. FIG. **48** illustrates a mobile phone or other relatively small device as the portable electronic device **1**. FIG. **49** illustrates a tablet or other relatively large device as the portable electronic device **1**. Many docking cradles can accommodate both large and small portable electronic devices, but other docking cradles may be specifically designed for a portable electronic device of a particular size or range of sizes.

[0187] FIG. **48** illustrates a cable **15** exiting the docking cradle **5**. It will be understood that the other illustrated docking cradles can include a cable. The cable may be detachable from the docking cradle or can be more permanently attached to the docking cradle. In some embodiments that incorporate a detachable cable, the docking cradle includes a port, such as a USB or microUSB port, that can receive an end of the cable.

[0188] In some embodiments, the docking cradle can include electronic components for providing power, data, or other signals to the portable electronic device from the cradle. In other embodiments, the docking cradle is a “pass-through” device where the docking cradle is simply an intermediary between the portable electronic device and another device, such as a charger, computer, other portable electronic device, or the like. In these embodiments, the cable exiting the docking cradle can be coupled to this other device and, at least in some instances, the docking cradle merely transmits signals between the electronic contacts of the docking cradle and the cable.

[0189] In some embodiments, a hard (or rigid) shell can be utilized with the flexible cover or skin to provide additional protection to electronic device disposed within the skin. FIGS. **50A** and **50B** are front and back views of a flexible cover, or skin, **100** (shaded for illustration purposes) and a hard (or rigid) shell **500**. The electronic device will be disposed within the flexible cover, or skin,

100 as described above. Any of the flexible covers described above can be used for the flexible cover, or skin, **100** with the hard shell **500** modified to fit the specific cover.

[0190] The hard shell **500** can be made of any hard or rigid plastic material, such as, for example, polycarbonate, or the like; carbon fiber; metal; leather; or the like; or any combination thereof. In at least some embodiments, the hard shell **500** includes a back panel **504** (FIG. 50B) and a skirt **506** (FIG. 50A) which, with the cover **100**, forms a mouth opening **112** that communicates with a cavity **108**. Mouth opening **112** is sized to receive electronic device into cavity **108**. In some embodiments, the back panel **504** may not cover the entire back portion of the cover **100**. In some embodiments, the skirt **506** may not completely surround the mouth opening **112**, but may only extend around a portion (for example, no more than 90%, 75%, 50%, 40%, or less) of the mouth opening.

[0191] In the illustrated embodiment, the hard shell **500** includes openings through which portions of the flexible cover, or skin, **100** are accessible or protrude. For example, the hard shell **500** includes an opening **514** through which a portion of the adapter **114** (such as the nesting feature **159** or the contactor **120**) of the cover **100** protrudes or is accessible for connection to a docking cradle or other device. In at least some embodiments, the hard shell **500** includes one or more button openings **598** to allow access to soft buttons **198** of the cover **100** (or buttons of the electronic device) or from which the soft buttons **198** of the cover **100** (or buttons of the electronic device) can protrude. In at least some embodiments, the hard shell **500** can include one or more corner openings **595** which allow corner portions **195** of the cover **100** to protrude to enhance corner protection using the flexibility of the cover **100**. Both the hard shell **500** and cover **100** can include openings **196** to allow access to a camera, a headphone jack, a switch, a speaker, a microphone, or the like of the electronic device.

[0192] The combination of the hard shell **500** and the flexible cover **100** can align and support the electronic device. In addition, the protruding portions of the flexible cover **100**, such as the protruding corner portions **195**, can act as soft bumpers to protect the electronic device.

[0193] In alternative embodiments, the electronic device can be inserted into a hard shell and then the device and hard shell can be inserted into any one of the protective covers **100** described above. In yet other embodiments, a hard shell can be provided as a removable (or permanent) liner for any of the protective covers **100** described above. The hard shells of these embodiments will include an opening through which the male plug **116** of the adapter **114** can protrude to allow connection to the input/output socket of the electronic device by the male plug **116** of the adapter **114**.

[0194] Instead of a docking cradle, the portable electronic device and the cover can be coupled to an external adapter. FIGS. 51A-51D illustrate a portable electronic device **1** disposed in a cover **100** (which can be any of the covers described herein) and an external adapter **600** with a wire or cable **615** extending from the external adapter. In FIG. 51A, the external adapter **600** is coupled to the adapter **114** of the cover **100**. In FIGS. 51B-51D, the external adapter **600** is uncoupled from the adapter **114** of the cover **100**. In at least some embodiments, the external adapter **600** only couples to the adapter **114** of the cover **100** and may only contact the cover **100** at the adapter **114** and, optionally, portions of the cover immediately adjacent to the adapter. The external adapter **600** may not have a tray.

[0195] The external adapter **600** has a housing **602** with a plurality of electrical contacts **611** (FIGS. 51C and 51D) for mating with the contacts **122** of the contactor **120** of the adapter **114** of the cover **100**. The electrical contacts **611** are electrically coupled to the electrical wire or cable **615**. In some embodiments, the electrical wire or cable **615** is detachable from the external adapter **600**. In some embodiments, the cable **615** is any cable that may be used with the portable electronic device and includes a plug which fits into a corresponding receptacle on the external adapter **600**.

[0196] In at least some embodiments, the adapter **114** and the external adapter **600** include the components of the magnetic attraction couple described above with respect to the docking cradle **5**. Alternatively or additionally, the external adapter **600** and adapter **114** can include other coupling

elements to reliably attach the external adapter **600** to the cover **100**. It will be understood that the docking cradles described above can also include any of these coupling elements.

[0197] In at least some embodiments, the external adapter **600** includes, as a coupling element, at least one ridged element **640** that can engage a detent **141** in the adapter **114**. In the illustrated embodiment of FIGS. **51A-51D**, the external adapter **600** includes two ridged elements **640** opposite each other and the adapter **114** includes two detents **141** (one of which is not shown) on opposite surfaces of the adapter **114**. In at least some embodiments, the ridged elements **640** are, prior to engagement of the adapter **114**, spaced apart by less than the corresponding width of the adapter **114** so that, when engaged in the detents **141**, the two ridged elements (and the external adapter **600**) form a compression fit with the adapter **114** that resists decoupling of the ridged elements **640** from the detents **141**.

[0198] In at least some embodiments, the external adapter **600** further includes a tab **642** extending from one of the ridged elements **640** (or two tabs with each tab extending from a different one of the ridged elements) in a cantilever or other arrangement so that a user can push on the tab to cause the ridged element to disengage from the detent **14** so that the external adapter **600** can be decoupled from the adapter **114** of the cover **100**. Other disengagement methods or mechanisms can also be used. It will be understood that in other embodiments, the cover may include at least one ridge element (and optional tab) and the external adapter may include at least one detent.

[0199] In at least some embodiments, the external adapter **600** can include one or more gripping members **644**, as a coupling element, that extend away from the remainder of the external adapter **600** and are arranged so that they form a compression or friction grip on the adapter **114** when the external adapter **600** engages the adapter **114**. This compression or friction grip resists decoupling of the external adapter **600** from the adapter **114**, but can be overcome by exertion of a decoupling force as the user pulls the external adapter **600** and adapter **114** apart. The illustrated embodiment of FIGS. **51A-51D** includes two opposing gripping members **644**, but it will be recognized that other embodiments can utilize more than two gripping members arranged around the external adapter **600** or a single gripping member that is formed around the entire (or a part of) the circumference of the external adapter **600**. In at least some embodiments, the gripping member(s) **644** can be shaped to fit the shape of the corresponding parts of the adapter **114**. For example, the adapter of FIG. **41** has a first edge **191** that is straight and a second edge **192**, opposite the first edge, that is curved and so gripping members **644** can be shaped with one having a shape to fit the straight first edge **191** and the other having a shape to fit the curved second edge **192**. It will be recognized that an external adapter (or docking cradle) can include one or both of the ridged elements **640** and gripping members **644**.

[0200] FIGS. **52A-F** illustrate one embodiment of a docking cradle **5** with a movable arm **43** to receive a portable electronic device **1** in a cover **100** (FIGS. **52E** and **52F**). In addition to the movable arm **43**, the docking cradle **5** includes a docking connector **3** with electrical contacts **11**, a tray **7**, base receiver **9**, and wire or cable **15** as described in any of the embodiments above. In at least some embodiments, the movable arm **43** includes a forward extension **45** that fits over an edge (for example, the top edge) of the portable electronic device **1** (FIGS. **52E** and **52F**) and can include a lip **47** that contacts the front surface of the portable electronic device **1** to assist in maintaining the portable electronic device engaged with the docking cradle.

[0201] The movable arm **43** can be moved from an extended position (FIGS. **52A** and **52B**) to a close position (FIGS. **52C** and **52D**) and any intermediate position, such as an engaged position (FIGS. **52E** and **52F**) where the movable arm **43** engages the portable electronic device **1** and cover **100** to and maintains the engagement of these components to the docking cradle **5**.

[0202] In the illustrated embodiment of FIGS. **52A-F**, the moveable arm **43** is coupled to a spring **750** (FIGS. **52B**, **52D**, **52F**) that is biased toward the close position of FIGS. **52C** and **52D**. It will be understood that other mechanical arrangements, such as a ratchet arrangement, can be used in place of the spring **750**. The spring **750** resists the inadvertent disengagement of the portable

electronic device **1** and cover **100** from the docking cradle **5** (as illustrated in FIGS. **52E** and **52F**). A user can extend the movable arm **43** away from the tray **7** to allow a portable electronic device to be placed on the tray followed by releasing the movable arm to allow the movable arm to engage the portable electronic device to hold the device on the docking cradle. The user can disengage the portable electronic device **1** and cover **100** from the docking cradle **5** by extending the movable arm away from the portable electronic device, cover, and tray **7** to release the portable electronic device and cover.

[0203] FIG. **53** illustrates another example of a docking cradle **5** similar to the docking cradles in FIGS. **48** and **49**. The docking cradle **5** includes a tray **7** and a base receiver **9** upon which a portable electronic device **1** within a cover **100** can be placed. The tray **7** includes a back support surface **101** to support a back of the cover **100** and a docking support surface **103** extending from the back support surface (also shown, for example, in FIGS. **3**, **4**, **37**, and **47**). In at least some embodiments, the back support surface **101** and the docking support surface **103** form an angle in a range of 90 to 130 degrees or a range of 90 to 120 degrees or in a range of 90 to 100 degrees or equal to 90 degrees.

[0204] The docking cradle **5** includes a docking connector **3** which, in the illustrated examples, is a female connector. The docking connector **3** includes contacts **11** which may be, for example, biasing contacts such as biasing pins or biasing leaf springs or any other suitable type of contact. The docking connector **3** also includes a rim **105** which may be recessed with respect to the docking support surface, as illustrated in FIG. **53**. In at least some embodiments, the rim **105** and docking connector **3** can be asymmetric, such as an elongated D-shape, as illustrated in FIG. **53**, with a straight side and a curved side opposite the straight side. Such an arrangement may facilitate proper alignment when the cover is mated with the docking cradle.

[0205] The docking cradle **5** of FIG. **53** is a stand-alone cradle that can be placed on any stable surface, such as a desk, shelf, credenza, table, or the like. FIGS. **54A-54B** illustrate another docking cradle **3** with multiple back support surfaces **101** and docking connectors **3** with one or more docking support surfaces **103** formed in a housing **107**. This can permit multiple electronic devices **1** in covers **100** to be docked in the docking cradle **3**. One or more cables (not shown) or a connector **109** can extend from the docking cradle for attachment to another device where the docking cradle is simply an intermediary between the portable electronic device and the other device, such as a charger, computer, other portable electronic device, or the like. In these embodiments, the cable or connector exiting the docking cradle can be coupled to this other device. It will be understood that the docking connector of any of the docking cradles disclosed herein can utilize the magnetic coupling arrangement described above and may also include one or more guide pins as described above.

[0206] FIG. **55A** illustrates a portion of an adapter **814** and includes a male plug **816**, electrical contacts **822**, magnets **878** (or ferromagnetic members), and a body **828**. The adapter **814** can be used in any of the covers described herein as an alternative to any of the adapters **114**. The adapter **814** is positioned within the cover **100** with the side skirt **116** and dam **132** formed around portions of the adapter, as illustrated in FIG. **55B**. In at least some embodiments, a portion of the shell **102** or dam **132** also covers a face **893** of the body **828** and the magnets **878** leaving a portion of the electrical contacts **822** exposed similar to the embodiments illustrated in FIGS. **39** and **41**.

[0207] As described above, the electrical contacts **822** are electrically coupled to the electrical connectors of the plug **816** by conductors (see, electrical conductors **126** described above). It has been found that available arrangements of plugs and the conductors are not generally suitable for the high pressure injection molding process used to form with shell **102** of the cover **100**. In such processes, the adapter **814** is an insert placed in the mold and the shell **102** of the cover **100** is molded around the adapter. It has been found that the plastic that forms the shell **102** may enter openings in the adapter **814** and can break or damage the conductors due to the high pressure of the molding process and may also fill the interior region, if present, of a connector and prevent mating

with the corresponding female connector of the portable electronic device.

[0208] Accordingly, the adapter **814** includes a sealed housing **875** disposed around the conductors **126** and from which the plug **816** extends. The sealed housing **875** can be formed by low pressure injection molding around the portion of the adapter containing the conductors to form a relatively tight seal with the body **828** and the plug **816** to reduce or prevent flow of material between these components during the later high pressure injection molding process that forms the shell **102**.

[0209] FIGS. **57A** and **57B** illustrate another embodiment of a cover **900** that can receive a portable electronic device **1** (FIG. **45**) and a peripheral device **50** (FIG. **44**) as an alternative to the embodiments of cover **100** illustrated in FIGS. **42-46**. Unless indicated otherwise, cover **900** can include any or all of the features described with respect to cover **100** illustrated in FIGS. **42-46** or any other cover **100** described herein.

[0210] The cover **900** includes a shell **902** with a center panel **904**, a side skirt **906**, a lip **910**, a mouth opening **912**, and an interior cavity **908** to receive the portable electronic device. In addition, the cover **900** includes a peripheral cavity **909** that forms a peripheral pocket for receiving the peripheral device **50**. The peripheral cavity **909** can be formed using the center panel **904** and optionally part of the side skirt **906** or any other portion of the cover **900**. The peripheral cavity **909** defines a device surface **905** and one or more side surfaces **907**. In the illustrated embodiment, the peripheral cavity **909** and interior cavity **908** are contiguous and the peripheral device **50** can be inserted into the peripheral cavity **909** through the mouth opening **912** and interior cavity **908**. As described above, in other embodiments, the peripheral device can be inserted through a different opening (i.e., a peripheral opening.) The peripheral opening can be positioned in the center panel **904** or along any one or more of the four sides of the side skirt **906**. Also, as described above, in some embodiments, the cover can receive more than one peripheral.

[0211] In addition to the male plug **916**, the cover **900** includes a peripheral plug **917**. The peripheral plug **917** fits into the input/output socket of the peripheral device. The peripheral plug **917** may be designed for a particular peripheral device or may be suitable for multiple different devices.

[0212] FIGS. **57A** and **57B** illustrate one embodiment of a portion of an adapter **914** that includes both the male plug **916** and the peripheral plug **917**, as well as a body **928**, electrical contacts **922**, magnets **978** (or ferromagnetic members), and a body **928**. The electrical contacts **922** and magnets **978** are attached to the body and the male plug **916** is coupled to the body with the conductors that couple the male plug **916** to the electrical contacts **922** passing through the body.

[0213] The adapter **914** also includes a bridge **929** which is attached to the body **928**. The bridge **929** contains a carrier **931** upon which the conductors (e.g., wires, conductive traces, or the like) that couple the electrical connectors of the peripheral plug **917** to the electrical contacts **922**. In some embodiments, the conductors of the bridge **929** are attached directly to portions of the electrical contacts **922** that are exposed through the body **928** or the conductors can be coupled to the electrical contacts **922** through conductive vias, wires, or traces that pass through the body **928**. In at least some embodiments, the carrier **931** is at least 0.4 mm thick in order to withstand the high pressure injection molding of the shell **902** of the cover **900**. Thinner carriers may break apart during the injection molding process. The peripheral plug **917** includes one or more electrical connectors that connect to contacts within the peripheral device. This arrangement can be used to provide power to the peripheral device via the adapter **914**. In at least some embodiments, the adapter **914** and associated conductors can be arranged so that the peripheral device provides power, data, or both to the portable electronic device through the adapter **914**. In at least some embodiments, the male plug **916** and the peripheral plug **917** can both include a sealed housing **975**, **977** as described above to prevent or reduce inflow of plastic material to the high injection molding process for forming the shell **902** of the cover **901** as described above.

[0214] While the preferred and additional alternative embodiments of the invention have been illustrated and described, it will be appreciated that various changes can be made therein without

departing from the spirit and scope of the invention. Therefore, it will be appreciated that various changes can be made therein without departing from the spirit and scope of the invention. Accordingly, the inventor makes the following claims.

Claims

1. A protective case for a portable electronic device, the protective case comprising: a center panel and a side skirt surrounding the center panel, the center panel comprising an exterior surface, wherein the center panel and the side skirt form an interior cavity therebetween, wherein the interior cavity is configured and arranged to receive the portable electronic device, wherein the protective case is configured and arranged to cover a back face of the portable electronic device with the center panel, cover at least a portion of each of a plurality of side faces of the portable electronic device with the side skirt, and to extend around a peripheral edge of a front face of the portable electronic device to secure the portable electronic device within the protective case; a male plug extending into the interior cavity of the protective case with or without the portable electronic device received in the interior cavity, wherein the male plug comprises a plurality of first contacts, wherein the male plug is positioned in the protective case for mating with a female socket of the portable electronic device when received in the protective case; a contactor attached to the side skirt and comprising a contactor surface exposed on a center portion of one exterior surface of the side skirt; and a plurality of second contacts disposed on the contactor surface, wherein at least one of the second contacts is electrically coupled to at least one of the first contacts of the male plug, wherein the protective case defines a plurality of recesses in or immediately adjacent to the contactor surface, wherein the plurality of second contacts is configured to mate with an external connector comprising a plurality of protruding structures complementary in shape to the plurality of recesses; wherein the male plug and the second contacts are configured for data transfer between the portable electronic device and an external device and for charging the portable electronic device using an external charging device.
2. The protective case of claim 1, wherein the contactor is embedded in the side skirt.
3. The protective case of claim 1, wherein the male plug is attached to the side skirt.
4. The protective case of claim 1, wherein the protective case is flexible.
5. The protective case of claim 1, wherein at least one of the recesses is defined in the contactor surface.
6. The protective case of claim 1, wherein the plurality of second contacts comprises at least six circular contacts.
7. The protective case of claim 1, wherein the contactor is centered on the one exterior surface of the side skirt.
8. A docking system comprising: the protective case of claim 1; and the external connector comprising a plurality of connector contacts configured to mate with the plurality of second contacts of the protective case.
9. The docking system of claim 8, wherein the connector contacts are spring-loaded pogo pin contacts.
10. A docking system, comprising: the protective case of claim 1; and a docking station comprising a base and a surface extending from the base configured to support a back of the protective case, wherein the base comprises a docking connector as the external connector; wherein the docking connector comprises a plurality of docking connector contacts configured to mate with the plurality of second contacts of the protective case.
11. The docking system of claim 10, wherein the docking connector comprises at least six of the docking connector contacts.
12. The docking system of claim 10, wherein the docking connector contacts are spring-loaded pogo pin contacts.

13. The docking system of claim 12, wherein the docking station is configured to facilitate transfer of power to or from the portable electronic device through the protective case and to facilitate transfer of data to or from the portable electronic device through the protective case.
14. The docking system of claim 13, wherein the docking connector is centered on the base of the docking station.
15. The docking stations of claim 14, wherein the base of the docking station comprises the protruding structures complementary in shape to the recesses of the protective case.
16. The docking system of claim 15, wherein the surface of the docking station extends from the base at approximately 90 degrees when the protective case is docked in the docking station.
17. The docking system of claim 10, wherein the docking station comprises a plurality of the docking connectors and a plurality of the surfaces extending from the base for supporting the backs of a plurality of the portable electronic devices, wherein the docking station is configured to charge the plurality of the portable electronic devices in the protective cases.
18. A protective case for a portable electronic device, the protective case comprising: a center panel and a side skirt surrounding the center panel, the side skirt comprising a plurality of interior and exterior side surfaces and the center panel comprising an exterior surface, wherein the center panel and the side skirt form an interior cavity therebetween, wherein the interior cavity is configured and arranged to receive the portable electronic device, wherein the protective case is configured and arranged to cover a back face of the portable electronic device with the center panel, cover at least a portion of each of a plurality of side faces of the portable electronic device with a different one of the interior side surfaces of the side skirt, and to extend around a peripheral edge of a front face of the portable electronic device to secure the portable electronic device within the protective case; a male plug extending into the interior cavity of the protective case with or without the portable electronic device received in the interior cavity, wherein the male plug comprises a plurality of first contacts, wherein the male plug is positioned in the protective case for mating with a female socket of the portable electronic device when received in the protective case; a contactor attached to the side skirt and comprising a contactor surface exposed on a center region of a first one of the exterior side surfaces of the side skirt; and a plurality of second contacts disposed on the contactor surface, wherein at least one of the second contacts is electrically coupled to at least one of the first contacts, wherein the protective case defines at least one recess adjacent to the contactor surface and closer to the contactor surface than a nearest intersection between the first one of the exterior side surfaces of the side skirt and a second one of the exterior side surfaces of the side skirt, wherein the plurality of second contacts is configured to mate with an external connector comprising a protruding structure complementary in shape to the at least one recess adjacent to the contactor surface; wherein the male plug and the second contacts are configured for data transfer between the portable electronic device and an external device and for charging the portable electronic device using an external charging device.
19. The protective case of claim 18, wherein the contactor is embedded in the side skirt.
20. The protective case of claim 18, wherein the male plug is attached to the side skirt.
21. The protective case of claim 18, wherein the protective case is flexible.
22. A docking system comprising: the protective case of claim 18; and the external connector comprising at least six spring-loaded pogo pin contacts configured to mate with the plurality of second contacts of the protective case.
23. A docking system, comprising: the protective case of claim 18; and a docking station comprising a base and a surface extending from the base configured to support a back of the protective case, wherein the base comprises a docking connector as the external connector; wherein the docking connector comprises a plurality of docking connector contacts configured to mate with the plurality of second contacts of the protective case.
24. The docking system of claim 23, wherein the docking connector comprises at least six of the docking connector contacts.

25. The docking system of claim 24, wherein the docking connector contacts are spring-loaded pogo pin contacts.

26. The docking system of claim 25, wherein the docking station is configured to facilitate transfer of power to or from the portable electronic device through the protective case and to facilitate transfer of data to or from the portable electronic device through the protective case.

27. The docking system of claim 26, wherein the docking connector is centered on the base of the docking station.

28. The docking stations of claim 27, wherein the base of the docking station comprises at least one of the protruding structure complementary in shape to the at least one recess on the protective case.

29. The docking system of claim 28, wherein the surface of the docking station extends from the base at approximately 90 degrees when the protective case is docked in the docking station.

30. The docking system of claim 23, wherein the docking station comprises a plurality of the docking connectors and a plurality of the surfaces extending from the base for supporting the backs of a plurality of the portable electronic devices, wherein the docking station is configured to charge the plurality of the portable electronic devices in the protective cases.
